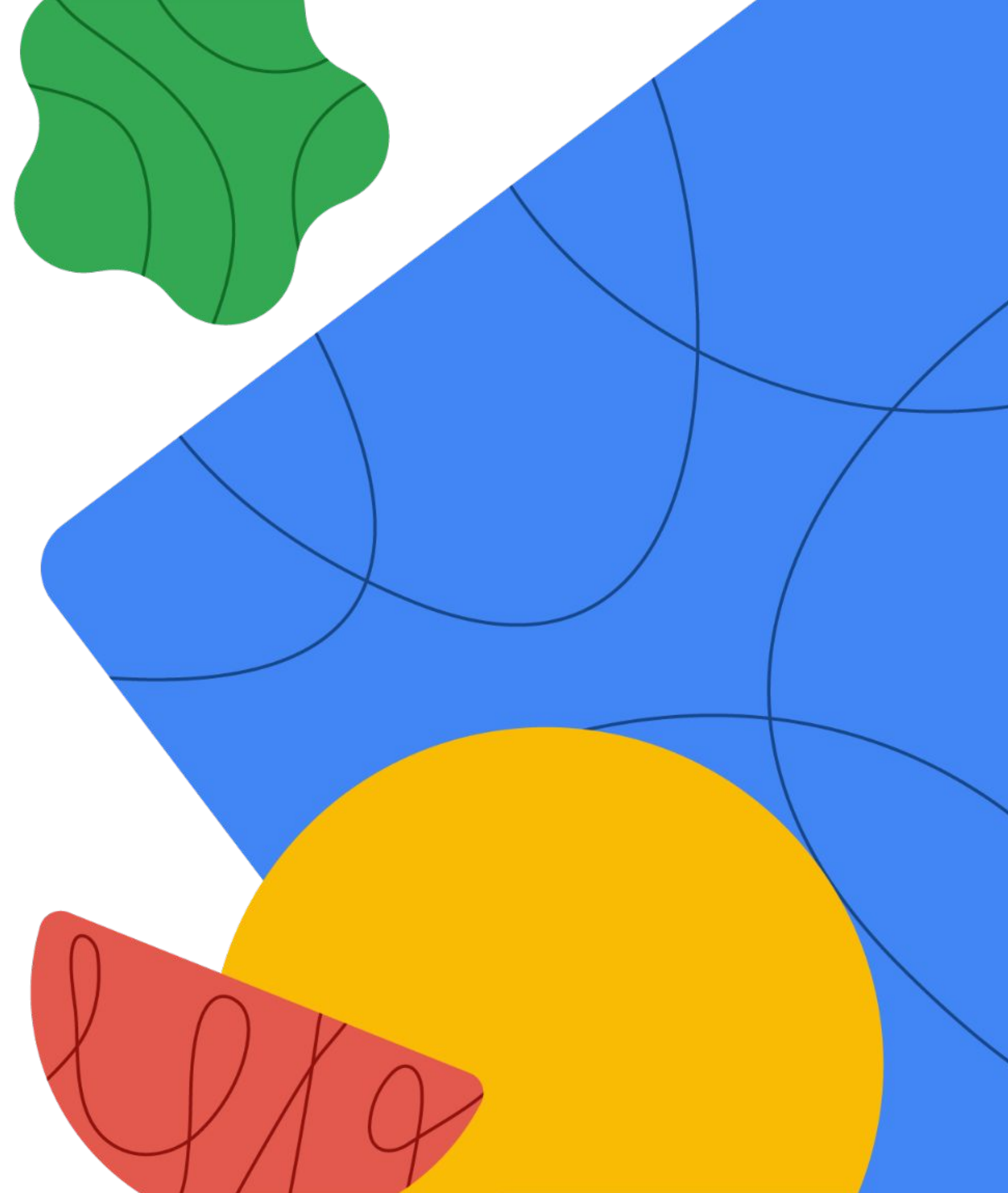


Video Info

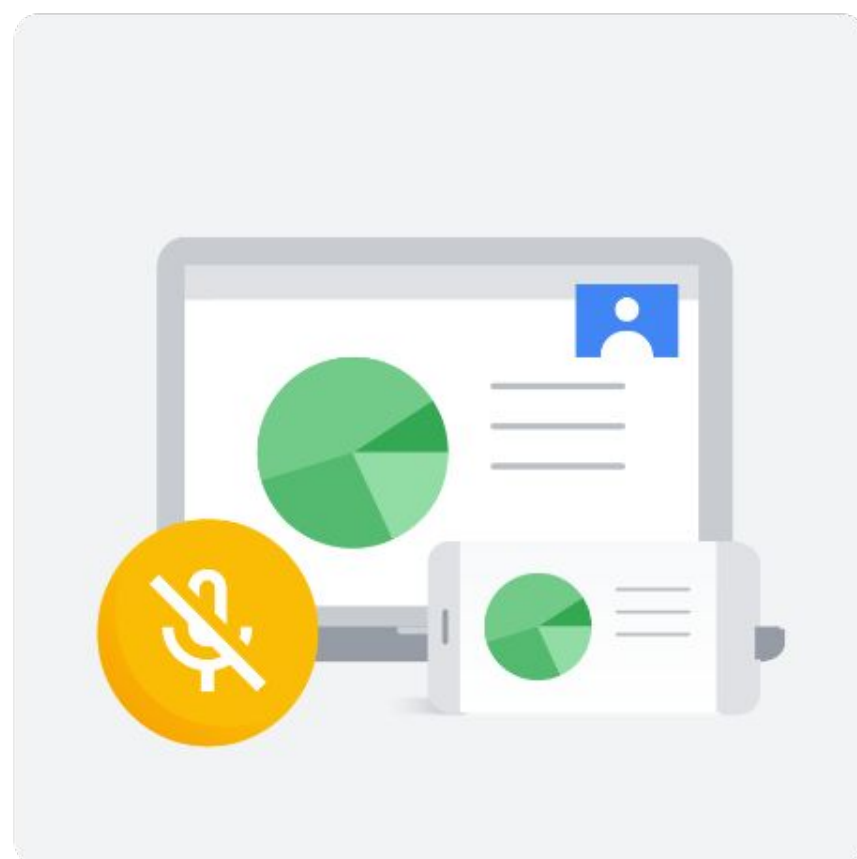
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Presenter:

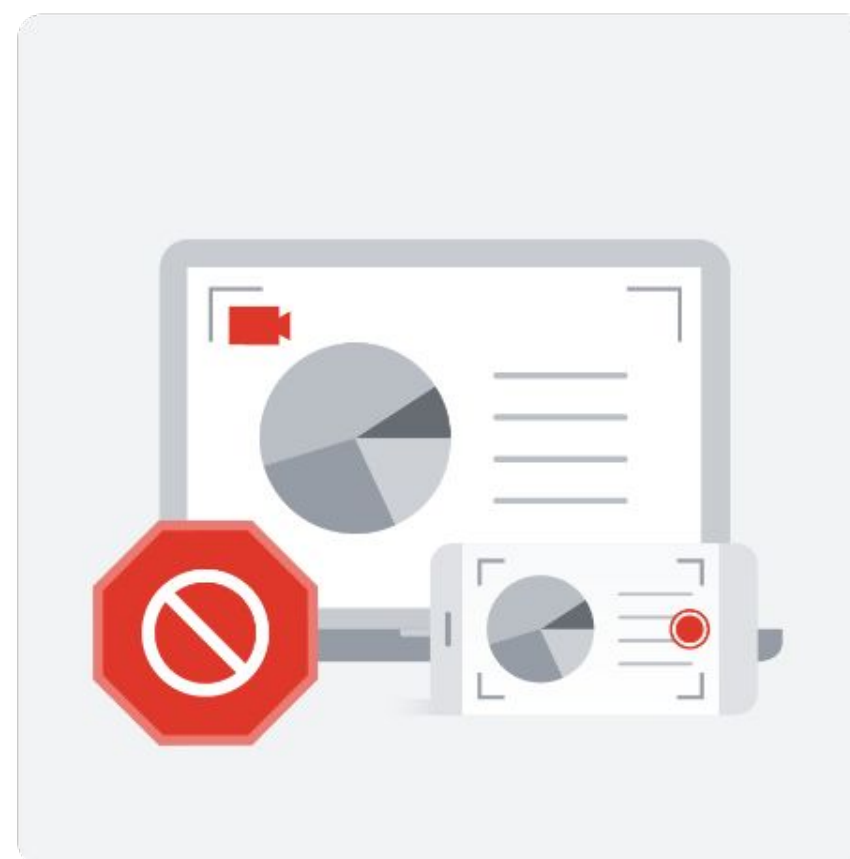
Google Cloud Digital Leader



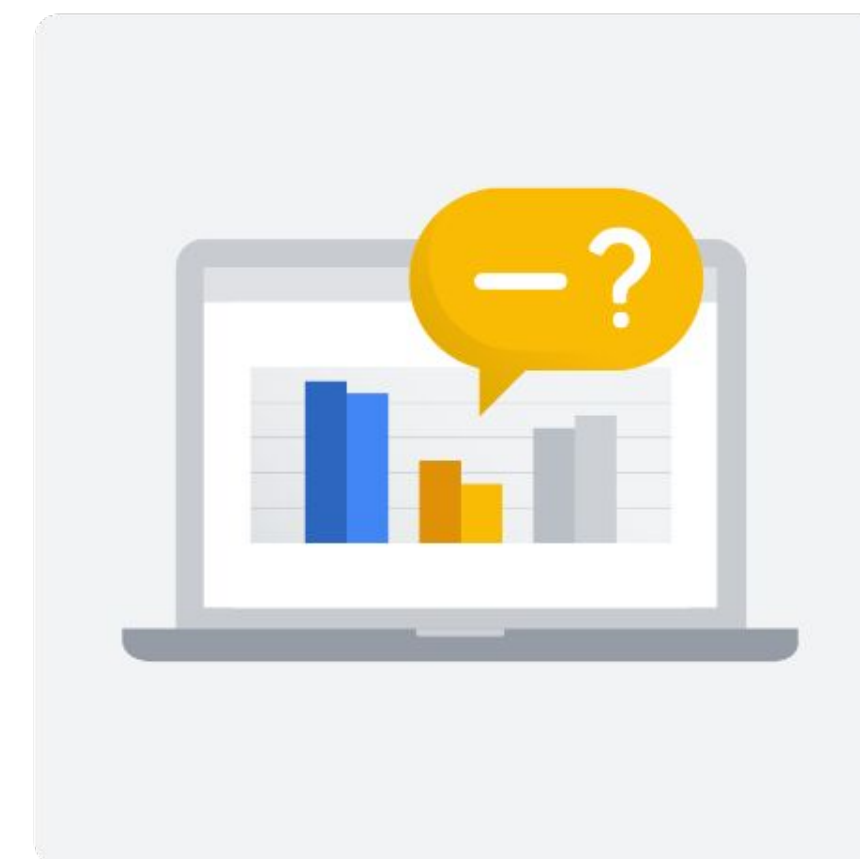
Etiquette



Mute microphone



No recording



Ask questions

Watch and Learn - CDL Training on Skills Boost

Google Cloud Skills Boost



Cloud Digital Leader Learning Path

This no cost learning path guides you through a curated collection of on-demand courses, labs, and skill badges that provide you with real-world, hands-on experience using Google Cloud technologies essential to the Cloud Digital Leader role.

📁 6 activities ⌚ Last updated 1 day 👤 Managed by Google Cloud

0% complete

A Cloud Digital Leader can articulate the capabilities of core cloud products and services and understand how they benefit organizations. This no cost learning path guides you through a curated collection of on-demand courses, labs, and skill badges that provide you with real-world, hands-on experience using Google Cloud technologies essential to the Cloud Digital Leader role. Once you complete the path, check out the Google Cloud Digital Leader certification (<https://cloud.google.com/certification/cloud-digital-leader>) to take the next steps on your professional journey.



01 Digital Transformation with Google Cloud

📁 Course ⌚ 8 hours ✓ Introductory +400 pts

There's much excitement about digital technology and digital transformation. But many unanswered questions remain. What is cloud technology? How can digital transformation mean business? How can technology help your business? If you're new to Google Cloud, you even begin? If you're an experienced professional, you'll find this course a great starting point for your journey.

→ Start course



02 Exploring Data Transformation with Google Cloud

📁 Course ✓ Introductory +400 pts



03 Innovating with Google Cloud Artificial Intelligence

📁 Course ⌚ 1 hour ✓ Introductory



04 Modernize Infrastructure and Applications with Google Cloud

📁 Course ✓ Introductory +400 pts

< 1% complete

Many traditional enterprises use legacy systems and applications that can't stay up-to-date with modern customer expectations. Business leaders often have to choose between maintaining their aging IT systems or investing in new products and services. 'Modernize Infrastructure and Applications with Google Cloud' explores the challenges and opportunities of modernizing your infrastructure and applications. This course provides a comprehensive overview of the Google Cloud ecosystem and the tools and services you need to modernize your infrastructure and applications. You'll learn how to design, build, and deploy modern applications on Google Cloud. You'll also learn how to manage and secure your infrastructure and applications. This course is a great starting point for your journey to modernize your infrastructure and applications.

Learn more



05 Trust and Security with Google Cloud

📁 Course ✓ Introductory +400 pts

As organizations move their data and applications to the cloud, they must address new security challenges. The Trust and Security with Google Cloud course explores the basics of cloud security, the value of Google Cloud's multilayered approach to infrastructure security, and the importance of security in the cloud. This course provides a comprehensive overview of the Google Cloud ecosystem and the tools and services you need to secure your infrastructure and applications. You'll learn how to design, build, and deploy secure applications on Google Cloud. You'll also learn how to manage and secure your infrastructure and applications. This course is a great starting point for your journey to secure your infrastructure and applications.

→ Start course



06 Scaling with Google Cloud Operations

📁 Course ⌚ 1 hour ✓ Introductory +400 pts

Organizations of all sizes are embracing the power and flexibility of the cloud to transform how they operate. However, managing and scaling cloud resources effectively can be a complex task. Scaling with Google Cloud Operations explores the fundamental concepts of cloud operations, including scaling, monitoring, and logging. This course provides a comprehensive overview of the Google Cloud ecosystem and the tools and services you need to scale your infrastructure and applications. You'll learn how to design, build, and deploy scalable applications on Google Cloud. You'll also learn how to manage and scale your infrastructure and applications. This course is a great starting point for your journey to scale your infrastructure and applications.

→ Start course

<https://partner.cloudskillsboost.google/paths/75>

Study Guide

Supplemental materials:

- Summary of each module
- Reference links
- Glossary of terms
- Glossary of products

[PDF](#)



Cloud Digital Leader certification exam

The Cloud Digital Leader certification helps you get up to speed with cloud concepts, so you can more effectively collaborate with your engineers and IT teams to achieve your organization's goals.

Length: **90 minutes**

Exam format: **Multiple choice and multiple select**

For more info visit:

cloud.google.com/certification/cloud-digital-leader



Certification Page

Cloud Digital Leader

A Cloud Digital Leader can articulate the capabilities of Google Cloud core products and services and how they benefit organizations. They can also describe common business use cases and how cloud solutions support an enterprise.

This certification is for anyone who wishes to demonstrate their knowledge of cloud computing basics and how Google Cloud products and services can be used to achieve an organization’s goals.

The Cloud Digital Leader exam assesses your knowledge in these areas:

- ✓ Digital transformation with Google Cloud
 - ✓ Innovating with Google Cloud artificial intelligence
 - ✓ Trust and security with Google Cloud
- ✓ Exploring data transformation with Google Cloud
 - ✓ Modernizing infrastructure and applications with Google Cloud
 - ✓ Scaling with Google Cloud operations

Quick links

[Train for the exam](#)

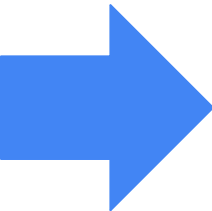
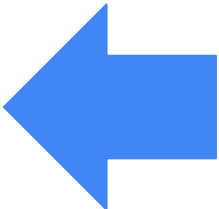
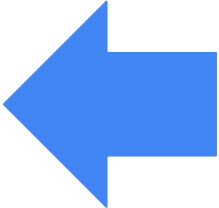
[Review sample questions](#)

[See Google Cloud Free Tier](#)

[Community: Learning & Certification Hub](#)

[Exam guide](#)

[Partner training for the exam](#)



Register

FAQs

[CDL Certification Exam Registration](#)

Modules

- 01 Digital Transformation with Google Cloud
- 02 Exploring Data Transformation with Google Cloud
- 03 Innovating with Google Cloud Artificial Intelligence
- 04 Modernize Infrastructure and Applications with Google Cloud
- 05 Trust and Security with Google Cloud
- 06 Scaling with Google Cloud Operations

Course agenda



Digital Transformation with Google Cloud

Module 1

Digital Transformation with Google Cloud



Lessons

- 01 Why cloud technology is transforming business
- 02 Fundamental cloud concepts
- 03 Cloud computing models and shared responsibility

Module 1

Digital Transformation with Google Cloud



Lessons

- 01 Why cloud technology is transforming business
- 02 Fundamental cloud concepts
- 03 Cloud computing models and shared responsibility

Cloud technology

is transforming how...

✓ Organizations create value

✓ People work

✓ People live

is the catalyst for thousands of innovations that change how...

✓ We navigate the world

✓ We interact with media

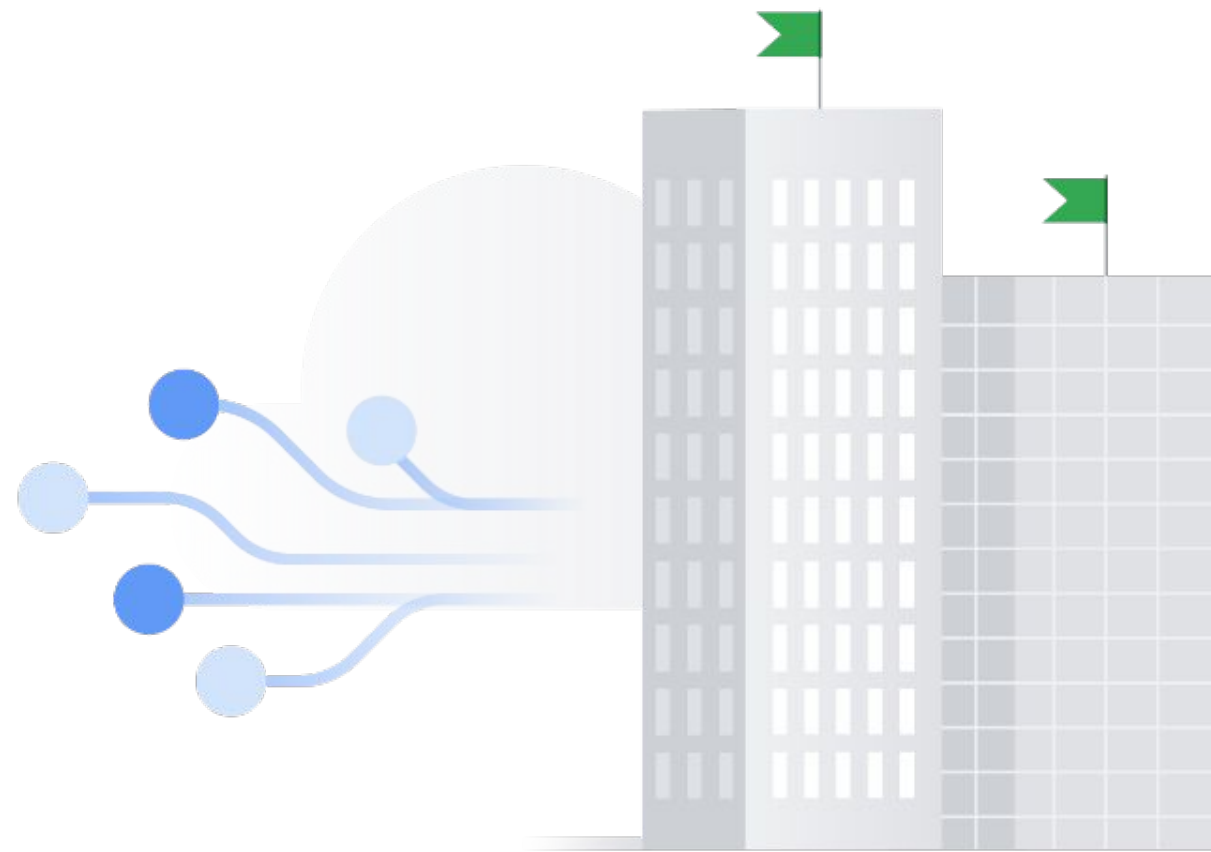
✓ We diagnose illness

✓ We combat environmental issues

Digital transformation

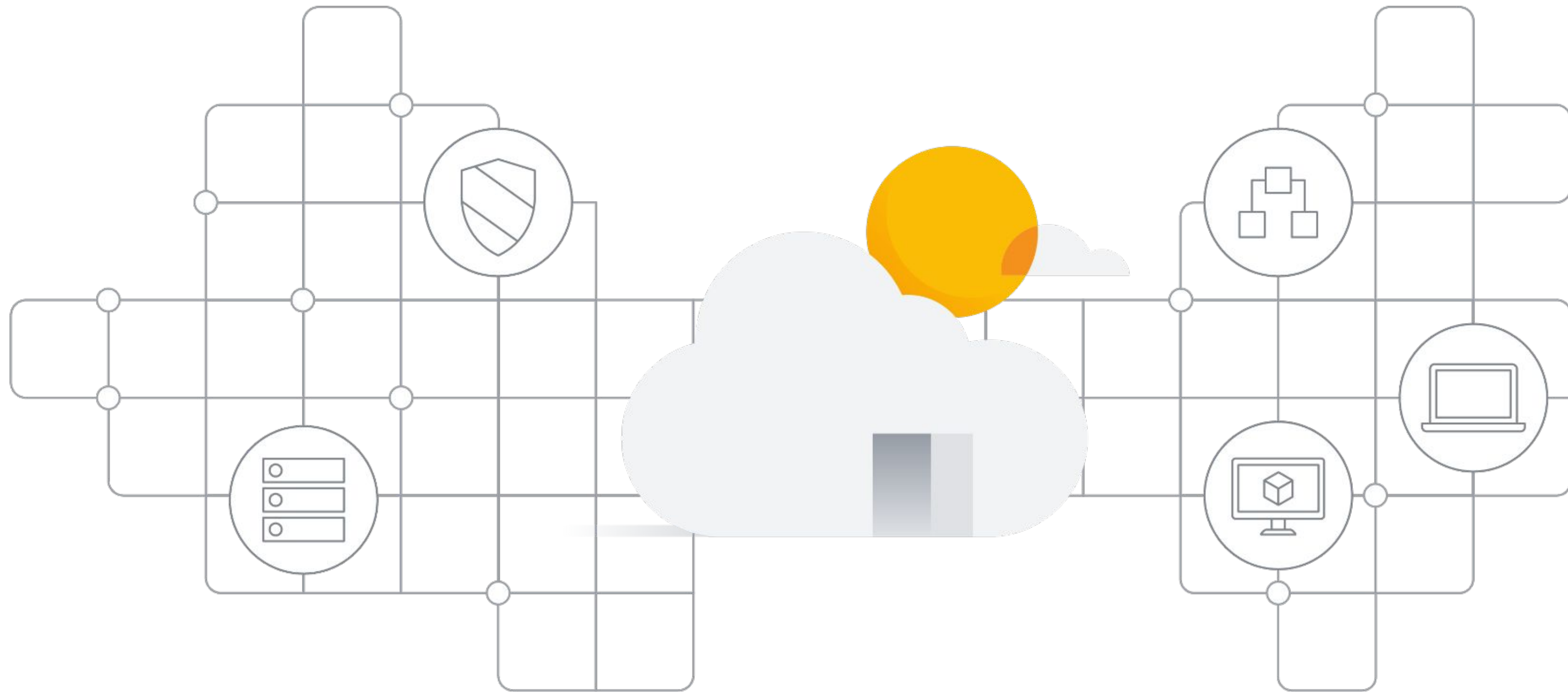
When an organization uses **new digital technologies**, such as public, private, and hybrid cloud platforms **to create or modify** business processes, culture, and customer experiences **to meet the needs** of changing business and market dynamics.

How does digital transformation help organizations?

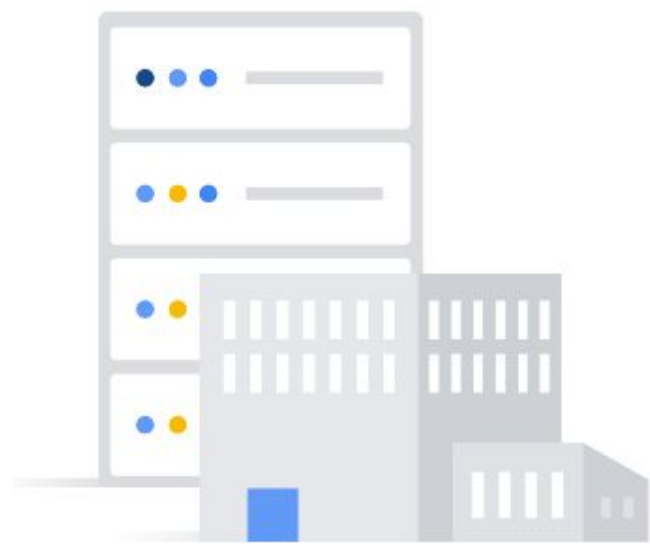


- ✓ Change how they operate
- ✓ Redefine relationships
- ✓ Modernize applications
- ✓ Create services
- ✓ Deliver value

The cloud is a metaphor for the network of data centers



IT infrastructure implementations



On-premises



Private cloud



Public cloud

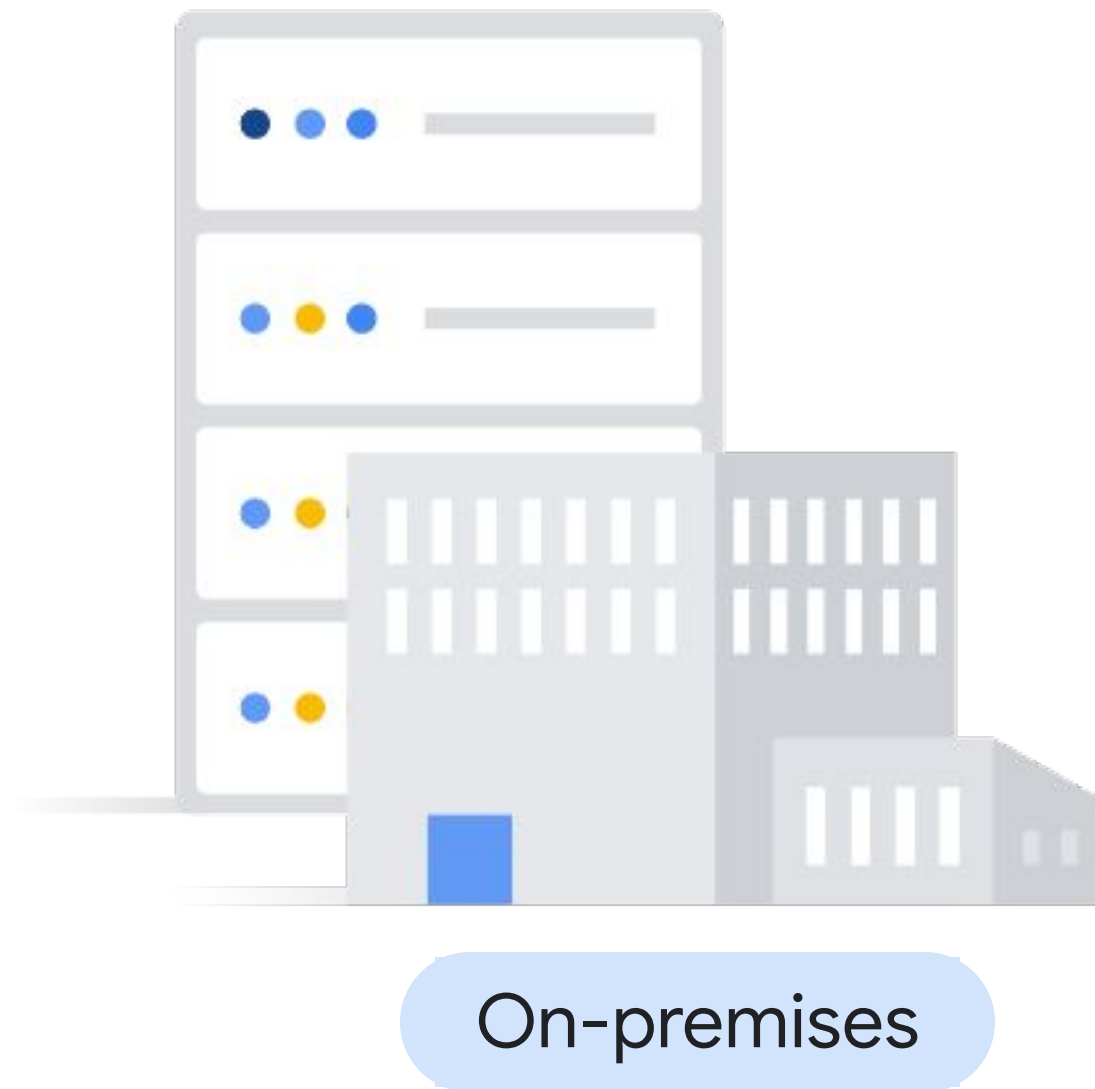


Hybrid cloud



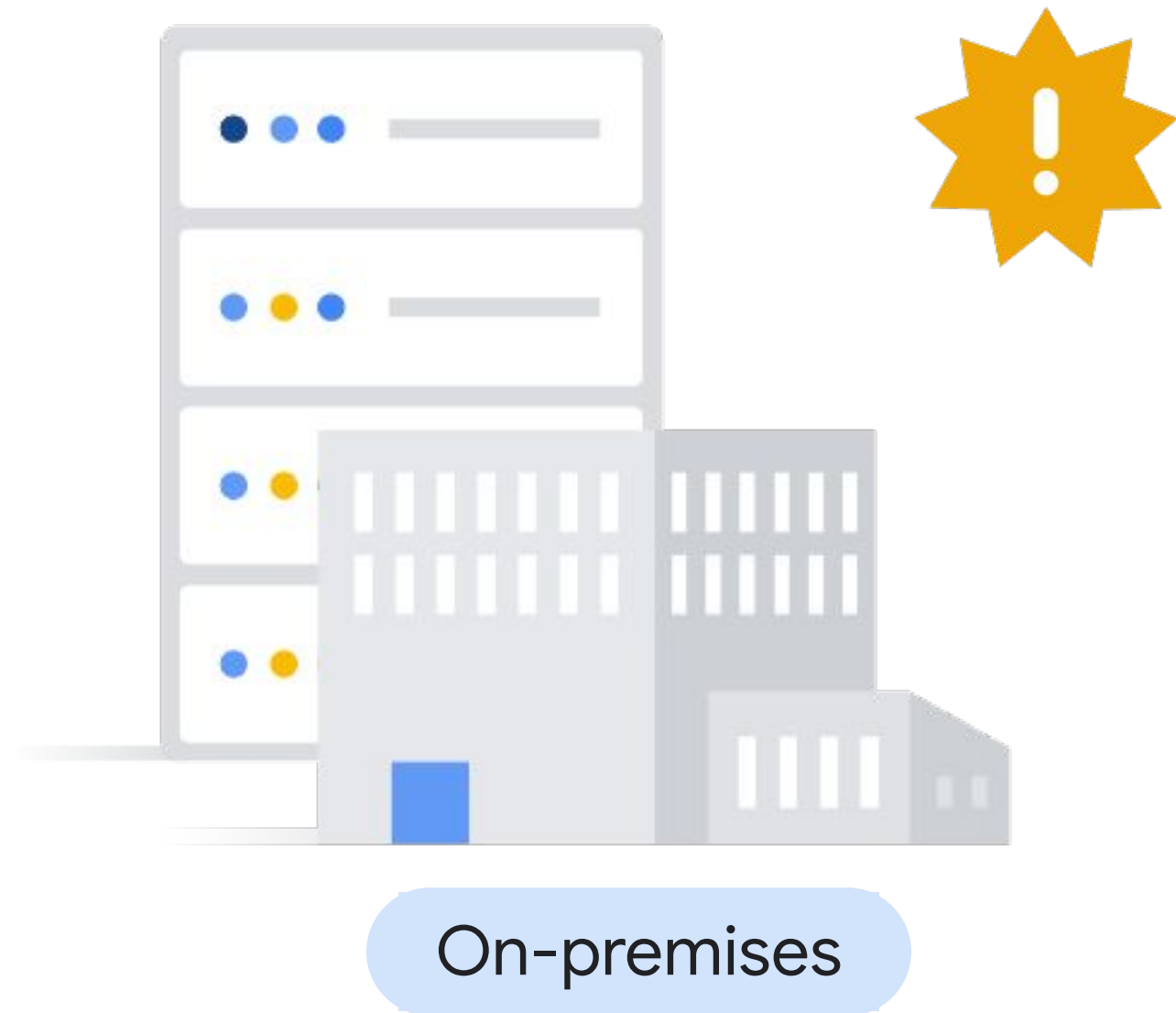
Multicloud

On-premises



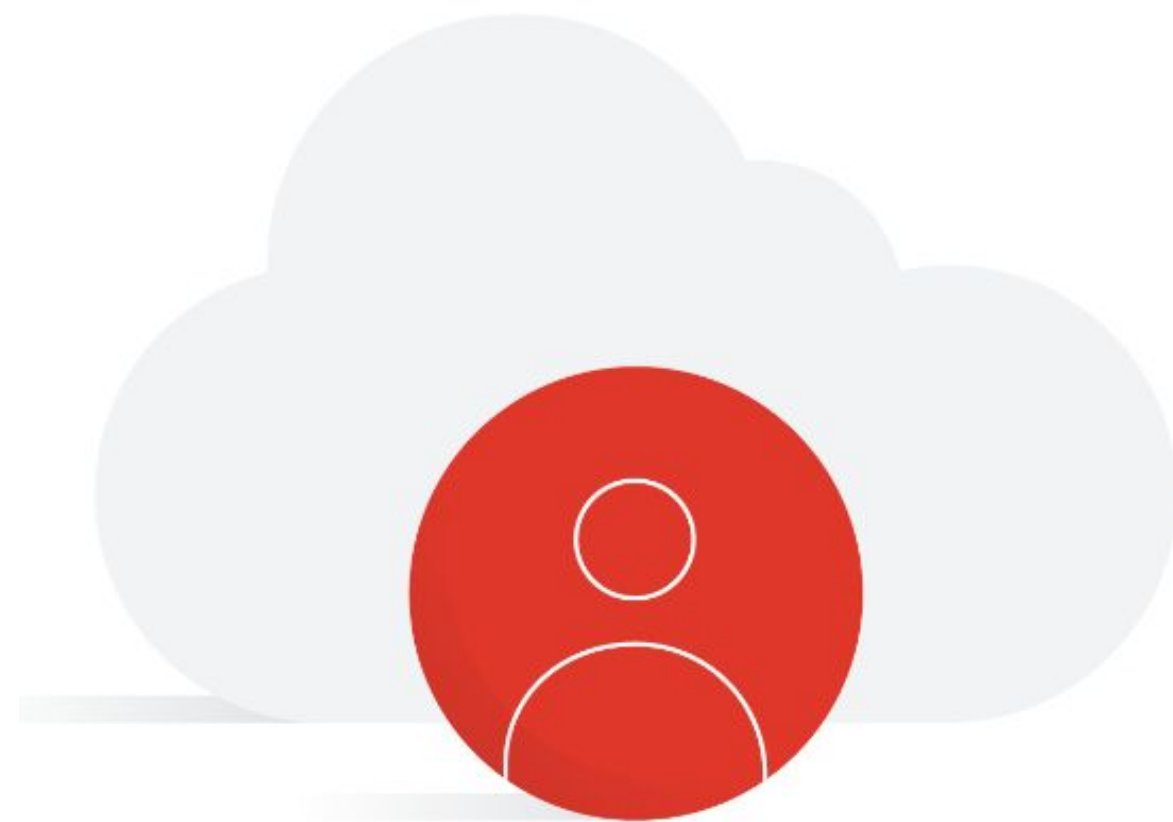
- Hosted on-site
- Located and operated in an organization's data center
- Traditional way of managing IT infrastructure
- Doesn't require third-party access
- Owners have physical control
- No payment for ongoing access

On-premises drawbacks



- Require physical space
- Require a specialized room
- Require expert personnel
- Difficult to scale
- Acquire more computing resources than needed

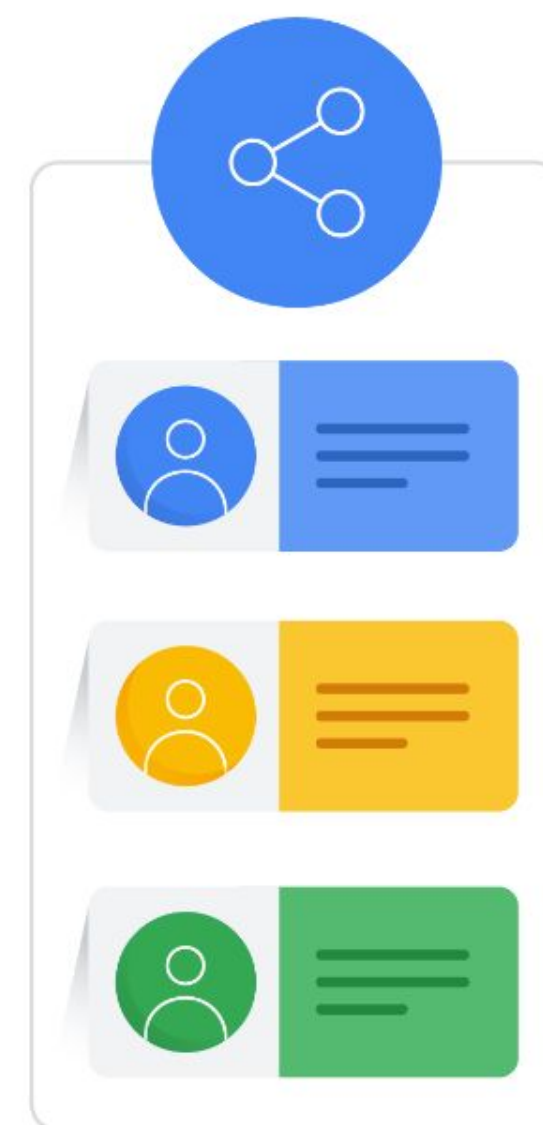
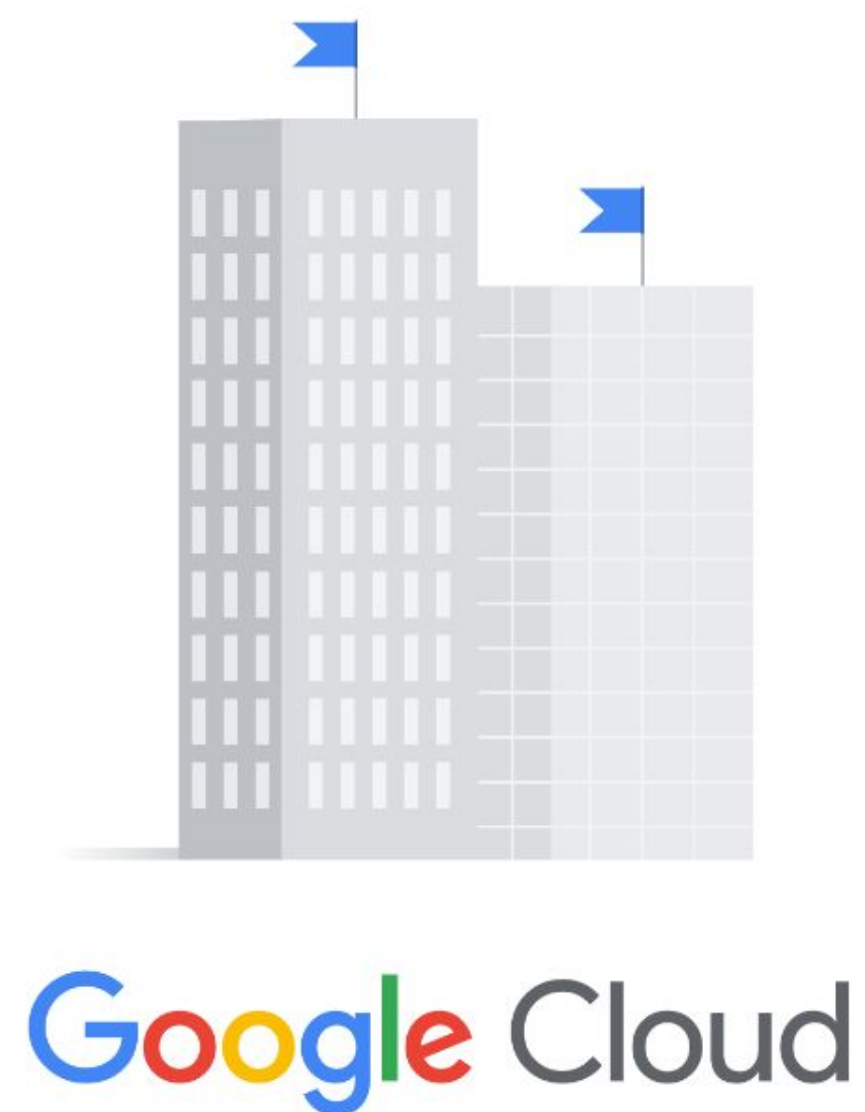
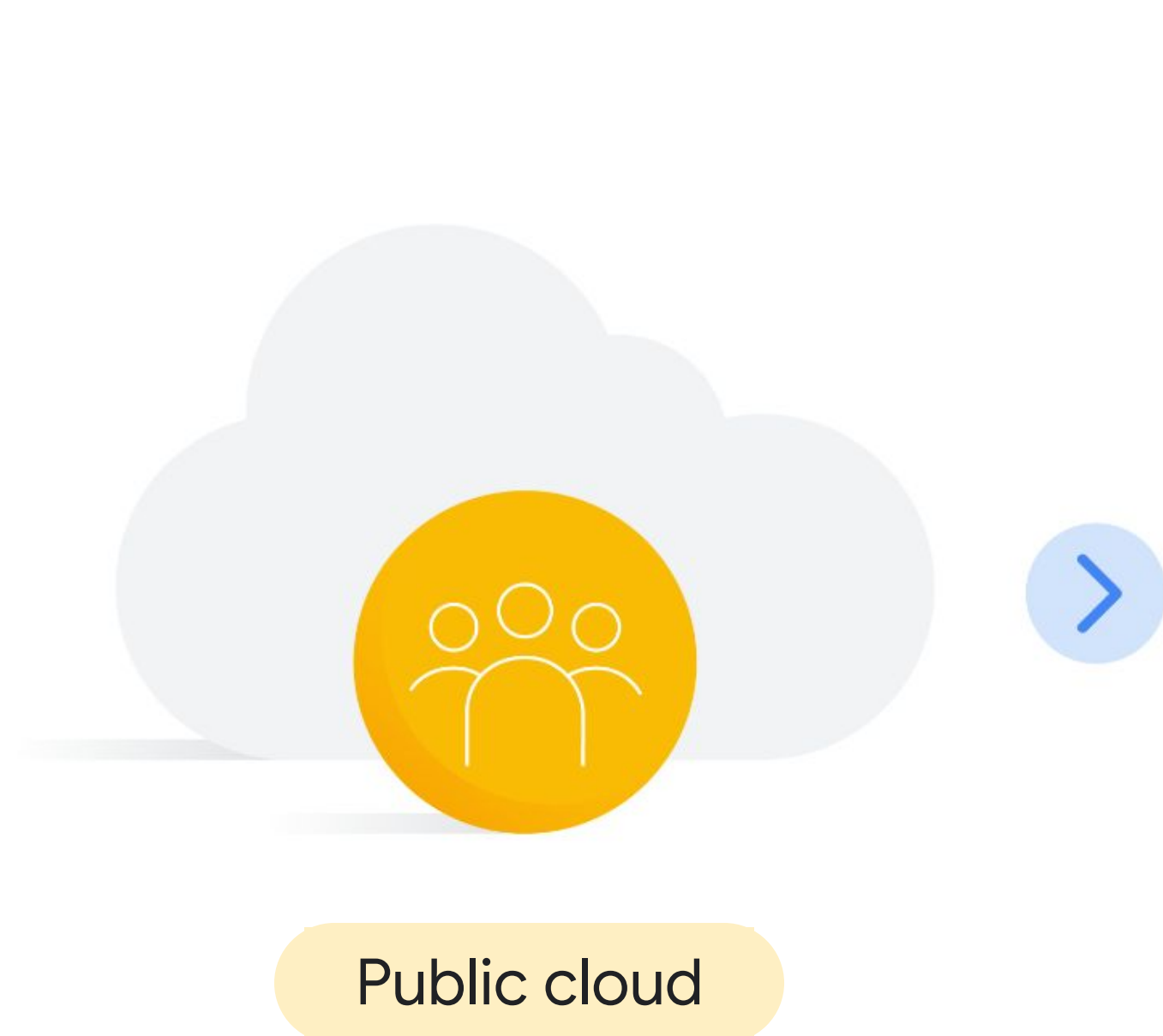
Private cloud



Private cloud

- Dedicated to a single organization
- Single-tenant or corporate cloud
- Same kind of ongoing maintenance and management as for traditional on-premises infrastructure
- Hosted within an organization's own private servers
- Benefits of a public cloud but with more customization available

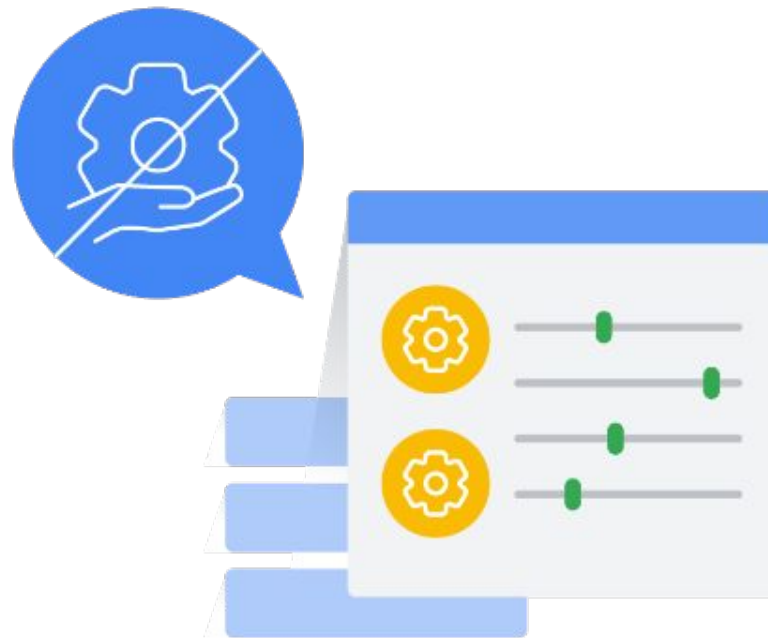
Public cloud



Public cloud benefits



On-demand availability
of computing and
infrastructure resources



No need to acquire,
configure, or manage
resources

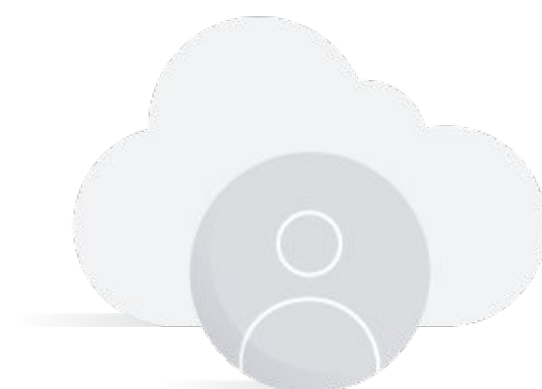


Pay only for
what is used

Hybrid cloud and multicloud



On-premises



Private cloud



Public cloud



Hybrid cloud

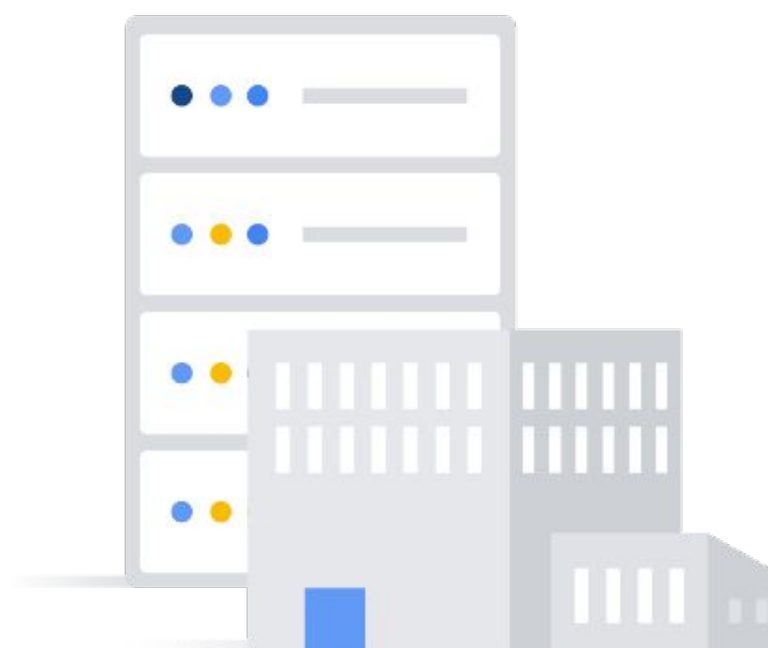


Multicloud

Hybrid cloud



Hybrid cloud

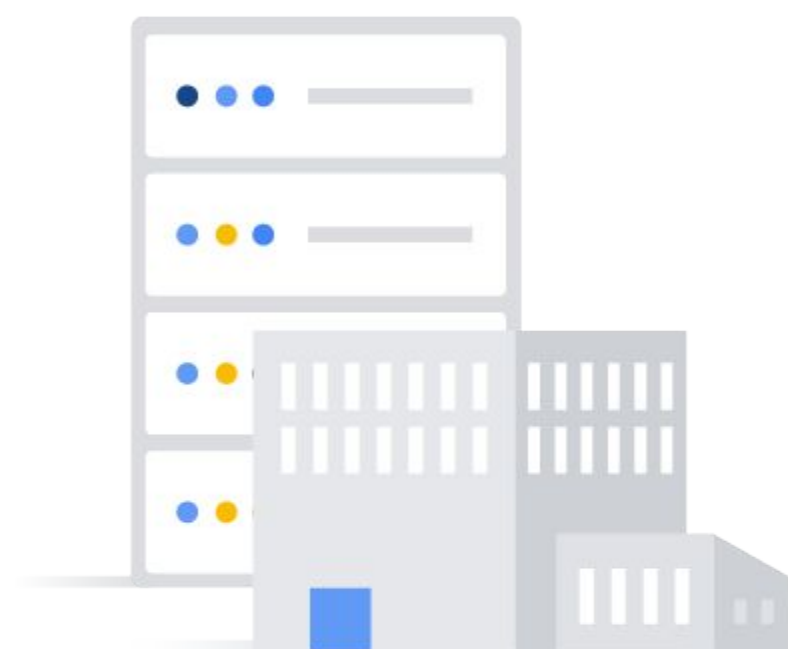


On-premises

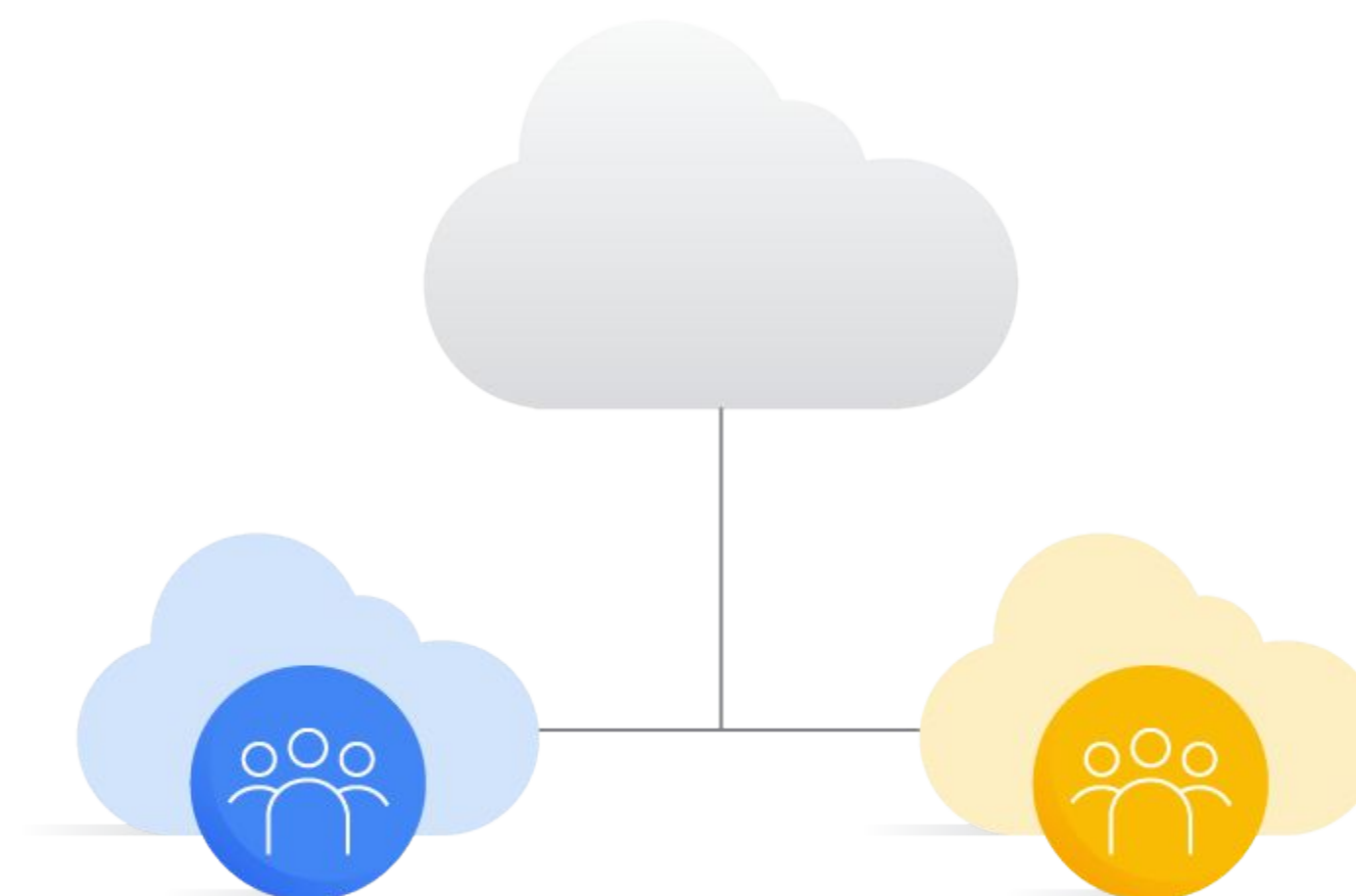


Google Cloud

Multicloud

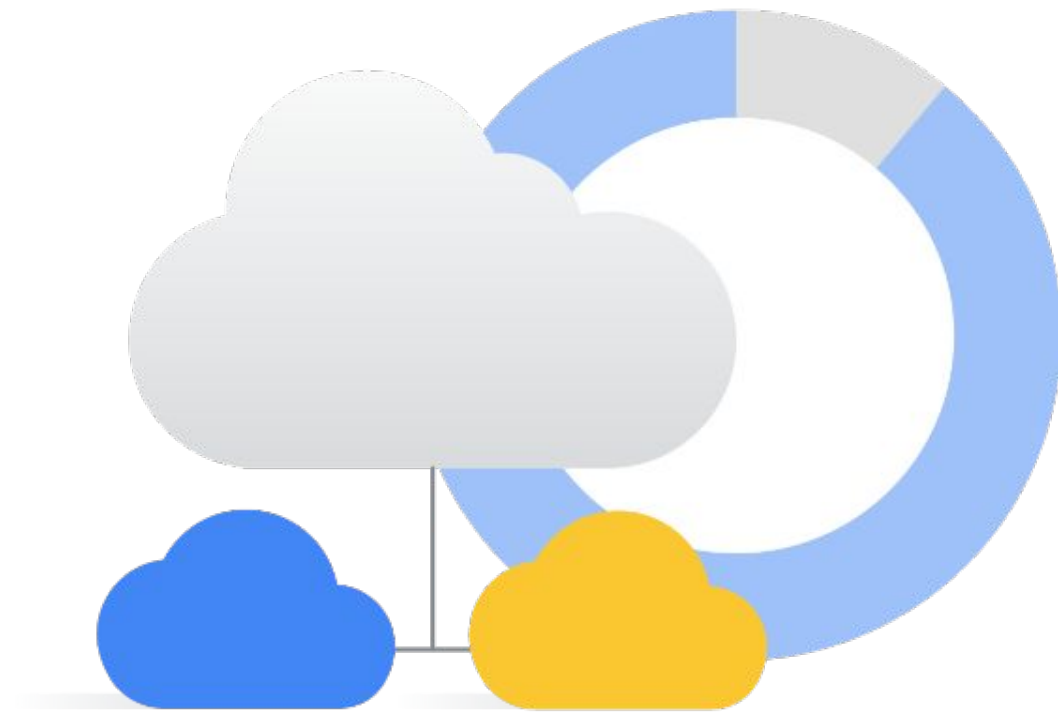


On-premises



Multiple public clouds

Today, most organizations embrace a multicloud strategy



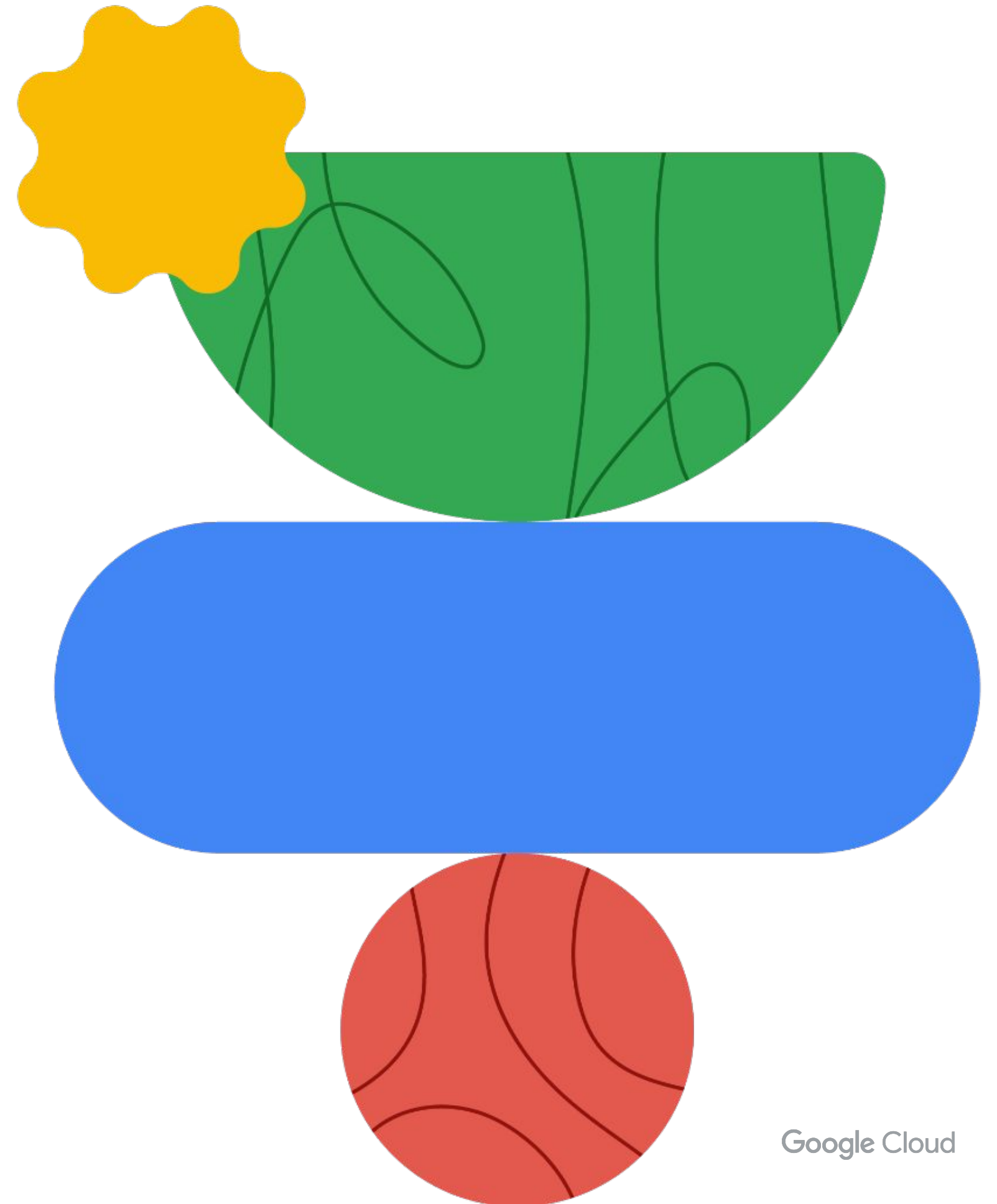
89% of respondents
have a multicloud strategy



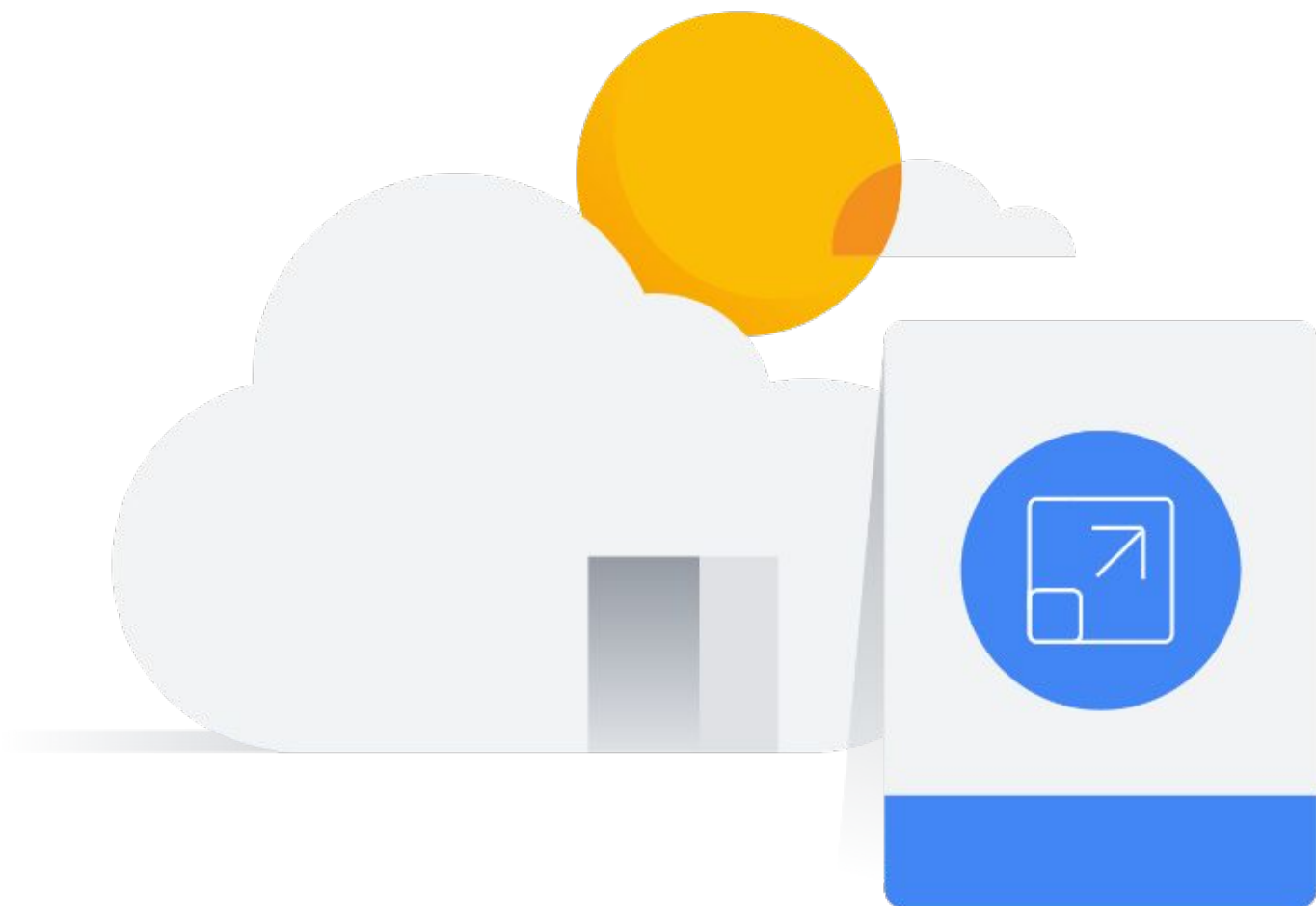
80% of respondents
have a hybrid strategy

Flexera 2022 State of the Cloud Report

The benefits of cloud computing

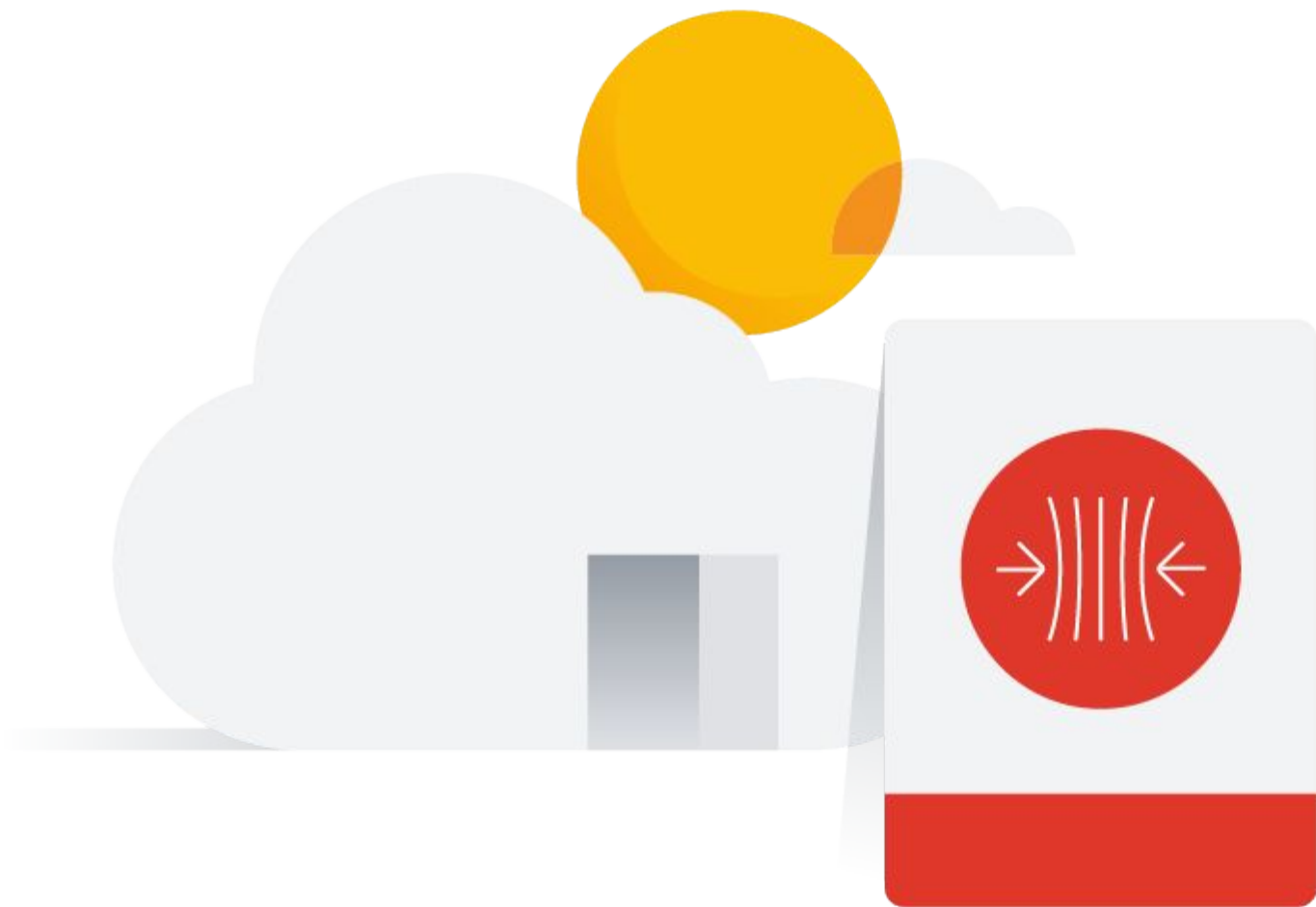


Scalability



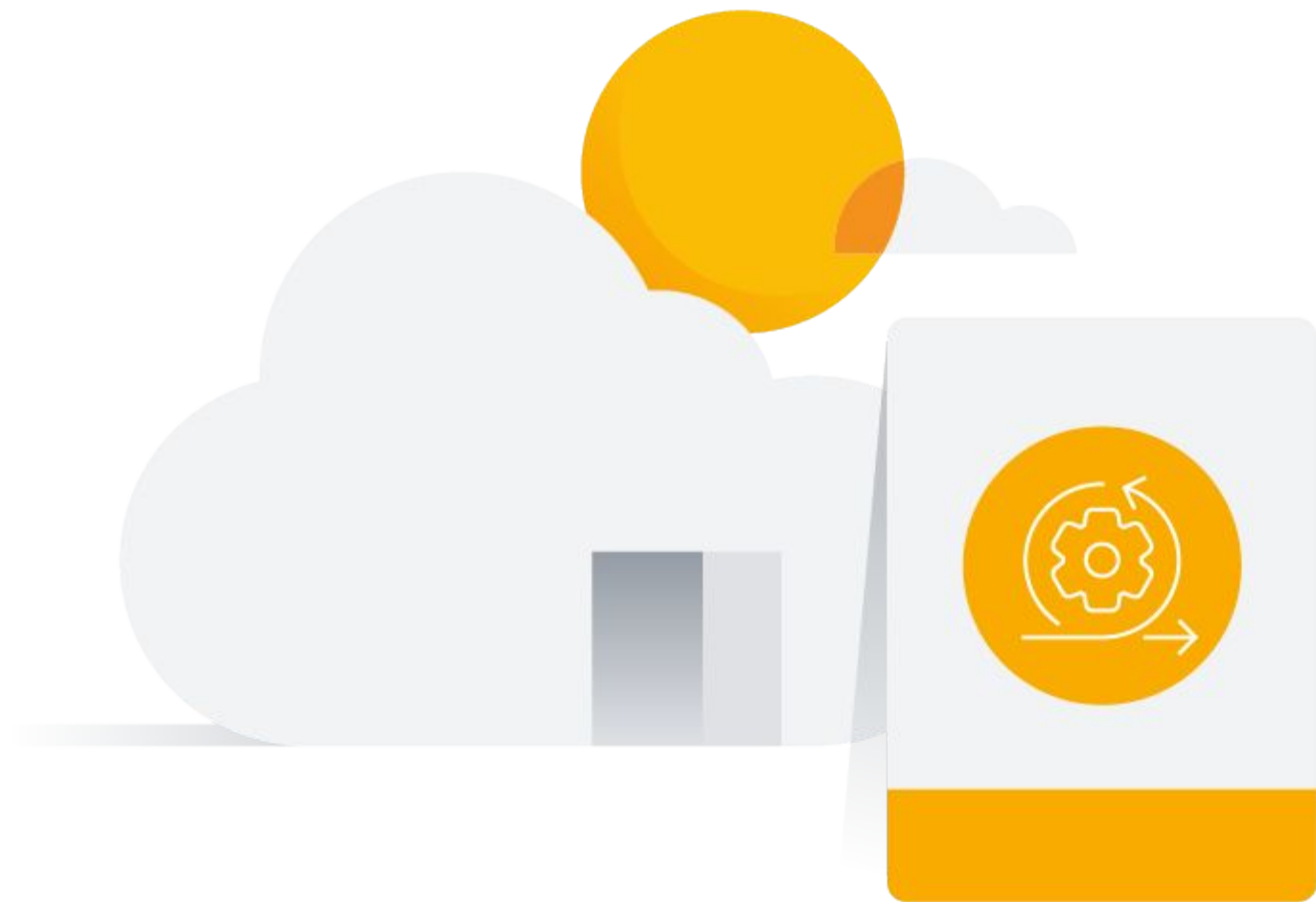
- ✓ Access to scalable resources
- ✓ Latest technology on-demand
- ✓ Accelerates deployment time

Flexibility



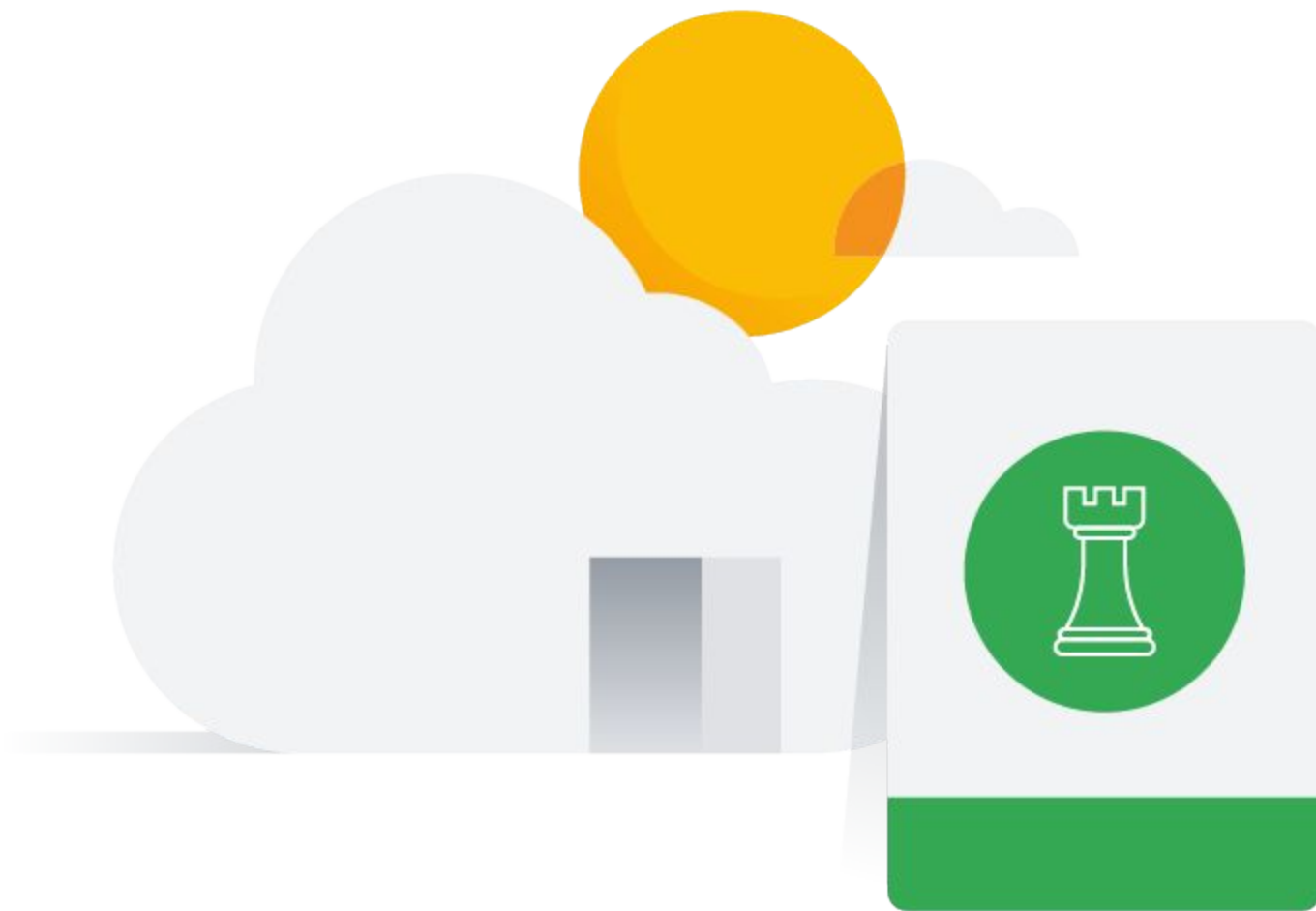
- ✓ Access services from anywhere
- ✓ Scale services up
- ✓ Scale services down

Agility



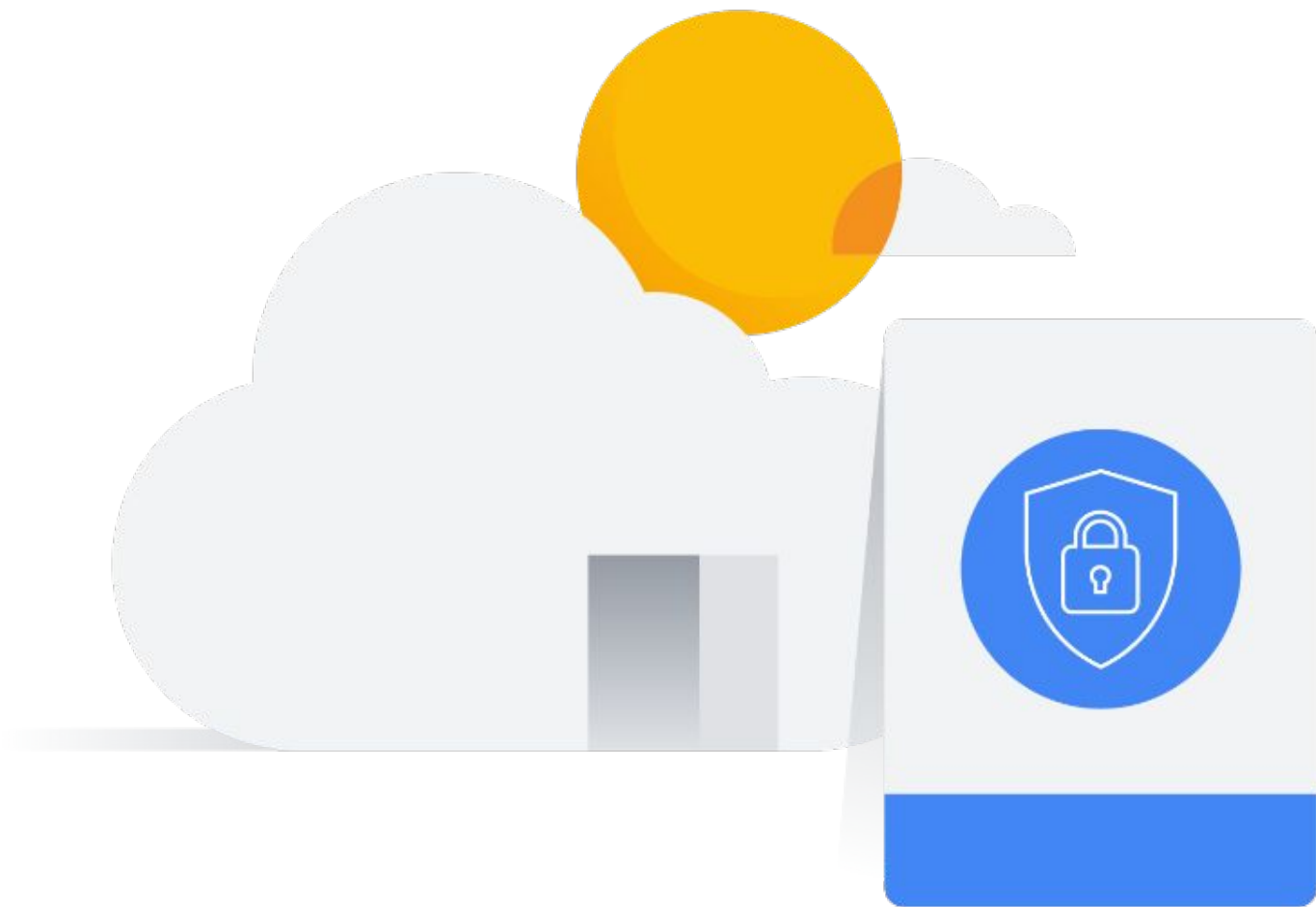
- ✓ Develop new applications
- ✓ Rapidly get them into production
- ✓ No infrastructure worries

Strategic value



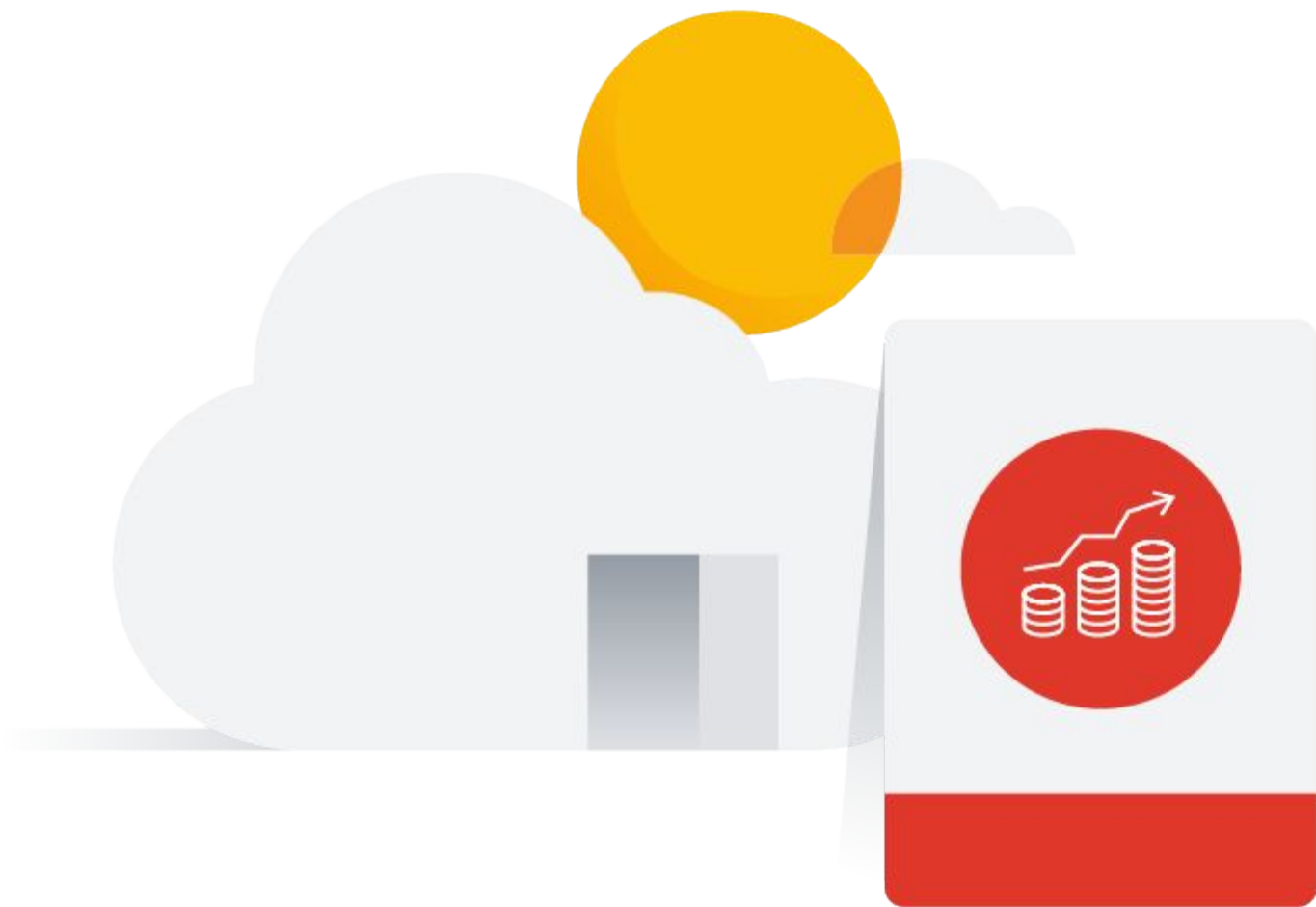
- ✓ Competitive advantages
- ✓ Higher return on investment
- ✓ Innovate and try new ideas

Security



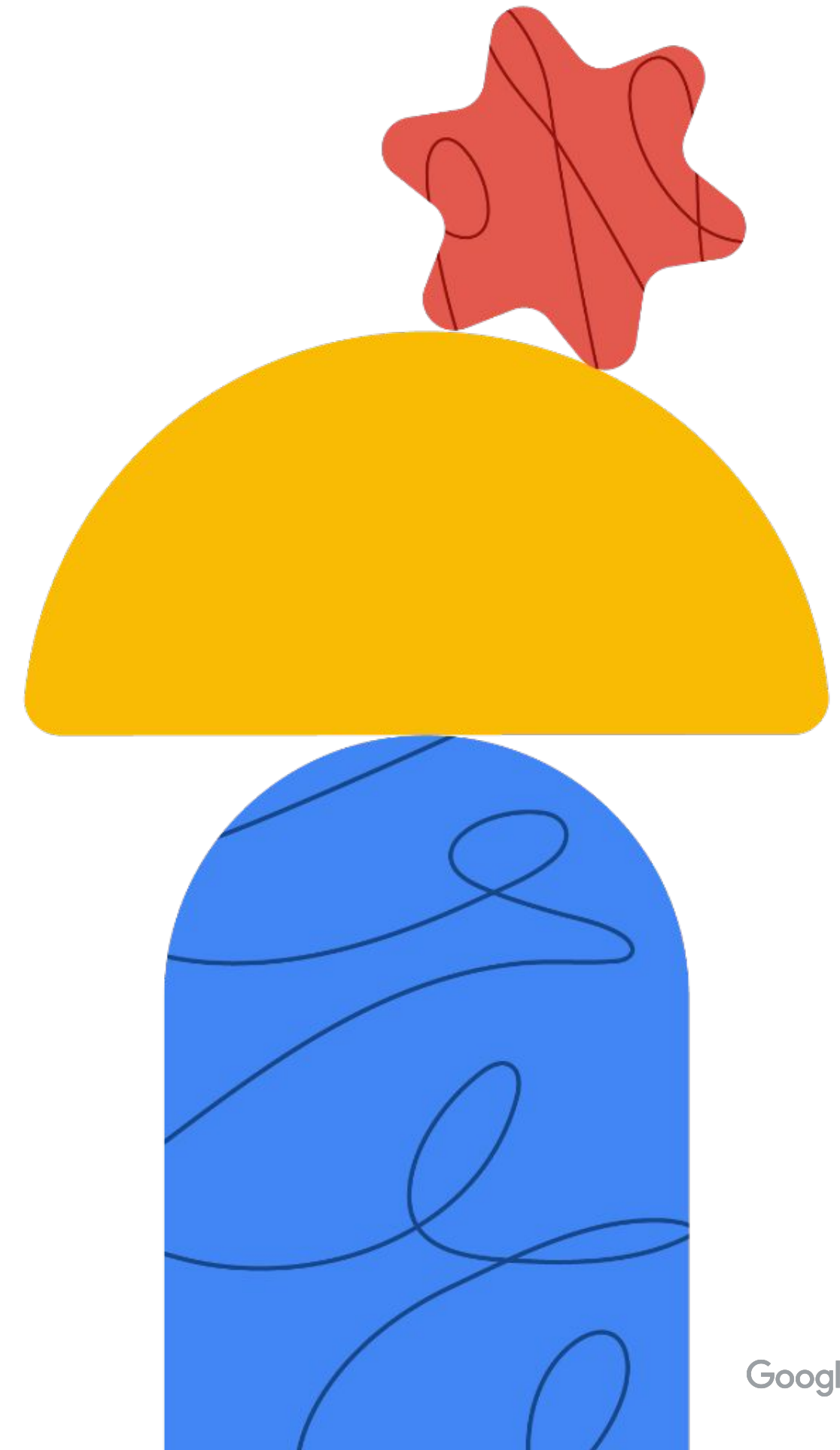
- ✓ Stronger than enterprise data centers
- ✓ Depth and breadth of mechanisms
- ✓ Dedicated teams

Cost-effectiveness



- ✓ Pay for what is used
- ✓ No overbuilding data centers
- ✓ IT staff can work on strategic initiatives

Google's transformation cloud



Capabilities that form the basis of the transformation cloud



Data



Open
infrastructure



Collaboration



Trust



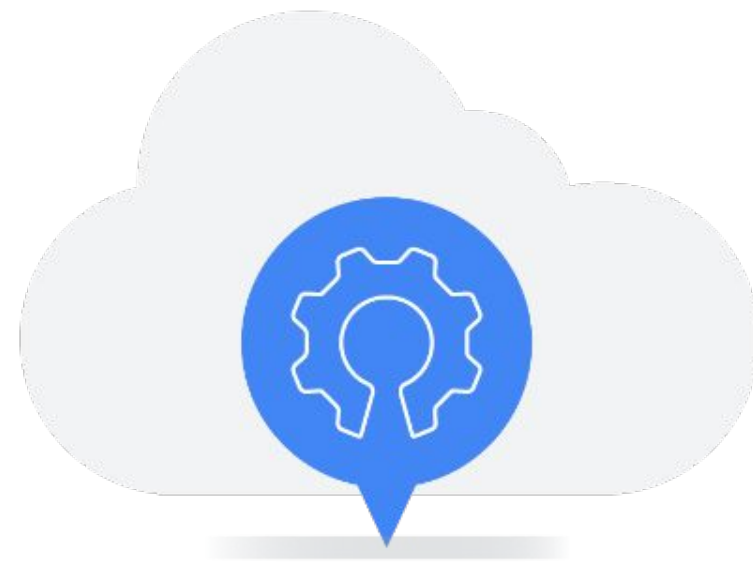
Sustainable
technology

Data cloud



- A data cloud is a unified solution to manage data across the entire data lifecycle.
- Data clouds let organizations identify and process data with great scale, speed, security, and reliability.
- Leading companies use a data cloud to encourage data-driven transformation, all with AI built in.

Open infrastructure cloud



Open
infrastructure

- Freedom to securely innovate and scale from on-premises, to edge, to cloud.
- Brings services to different physical locations, while leaving the operation, governance, and evolution of the services to Google Cloud.
- Run applications in the place that makes the most sense, using hybrid and multicloud approaches based on open-source software.
- Facilitates faster innovation and reduces lock-in to a single cloud provider.

Collaboration cloud



Collaboration

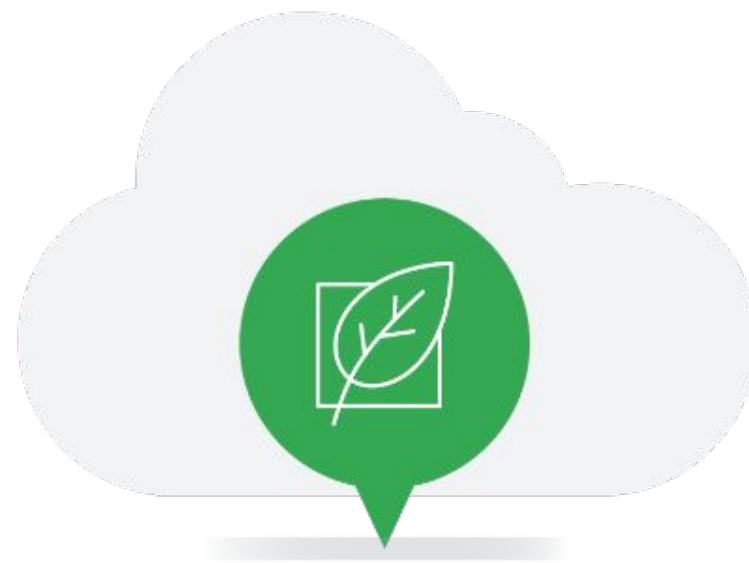
- Helps transform how people connect, create, and collaborate.
- People and culture are as important as technology.
- Organizations have increased both location and time flexibility in work arrangements.
- Information and frontline workers across regions and industries need to collaborate securely from anywhere, and on any device.

Trusted cloud



- A trusted cloud helps organizations protect what's important with advanced security tools.
- The annual cost of cyber crime is expected to reach **\$10.5 trillion** annually by 2025.
- Organizations are finding ways to identify and protect everything, from people and customers to data and transactions.
- Organizations want to find, analyze, resist, and remediate threats at global scale while maintaining control of their digital assets.

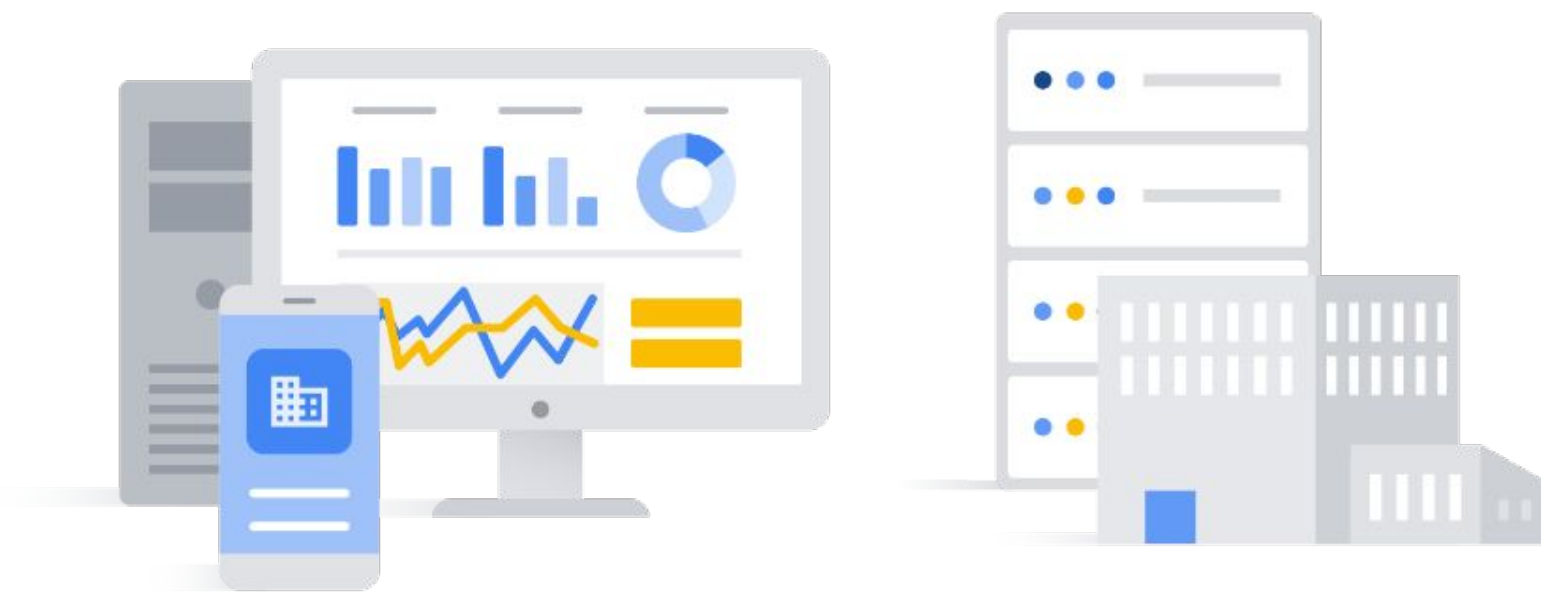
Sustainable technology



Sustainable
technology

- Using technology and solutions that help organizations build and work more sustainably.
- Cloud computing is estimated to save **1 billion metric tons** of CO2 emissions by 2024.
- The largest corporations have the opportunity to lead the way in helping the world reduce its emissions and operate on carbon-free energy.
- Google Cloud partners with customers to decarbonize their digital apps and infrastructure with our sustainable technology and solutions.

Total cost of ownership (TCO)



Dominated by the
initial purchase
of hardware and software

On-premises

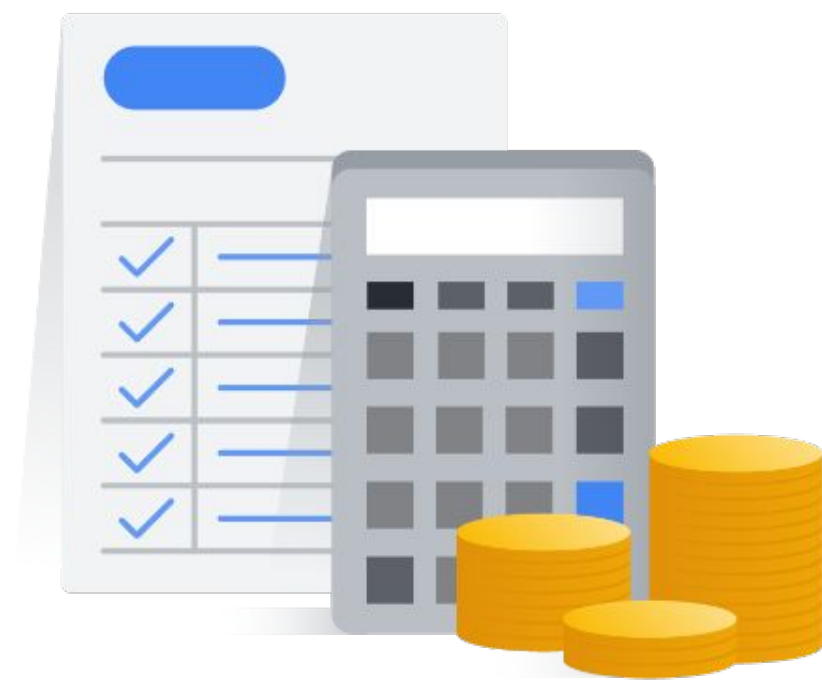


Cloud

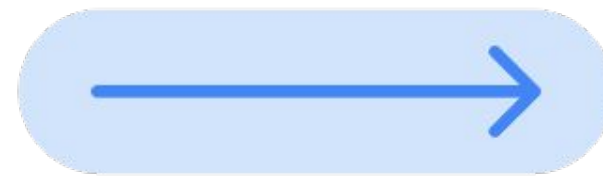


Based on monthly
subscriptions or
pay-per-use models

From capital expenditure to operating expenses



Capital
expenditures



Operating
expenses

Quiz

Question

An organization has shifted from a CapEx to OpEx based spending model. Which of these statements is true?

- A. They will only pay for what they forecast.
- B. Budgeting will only happen on an annual basis.
- C. They will only pay for what they use.
- D. Hardware procurement is done by a centralized team.

Quiz

Answer

An organization has shifted from a CapEx to OpEx based spending model. Which of these statements is true?

- A. They will only pay for what they forecast.
- B. Budgeting will only happen on an annual basis.
- C. They will only pay for what they use.
- D. Hardware procurement is done by a centralized team.



Quiz

Question

An organization has made significant investments in their own infrastructure and has regulatory requirements for their data to be hosted on-premises.
Which cloud implementation would best suit their needs?

- A. Public Cloud
- B. Private Cloud
- C. Platform as a service
- D. Software as a service

Quiz

Answer

An organization has made significant investments in their own infrastructure and has regulatory requirements for their data to be hosted on-premises.
Which cloud implementation would best suit their needs?

- A. Public Cloud
- B. Private Cloud
- C. Platform as a service
- D. Software as a service



Quiz

Question

An organization wants to innovate using the latest technologies, but also has compliance needs that specify data must be stored in specific locations. Which cloud approach would best suit their needs?

- A. Public Cloud
- B. Multicloud
- C. On-premises infrastructure
- D. Hybrid Cloud

Quiz

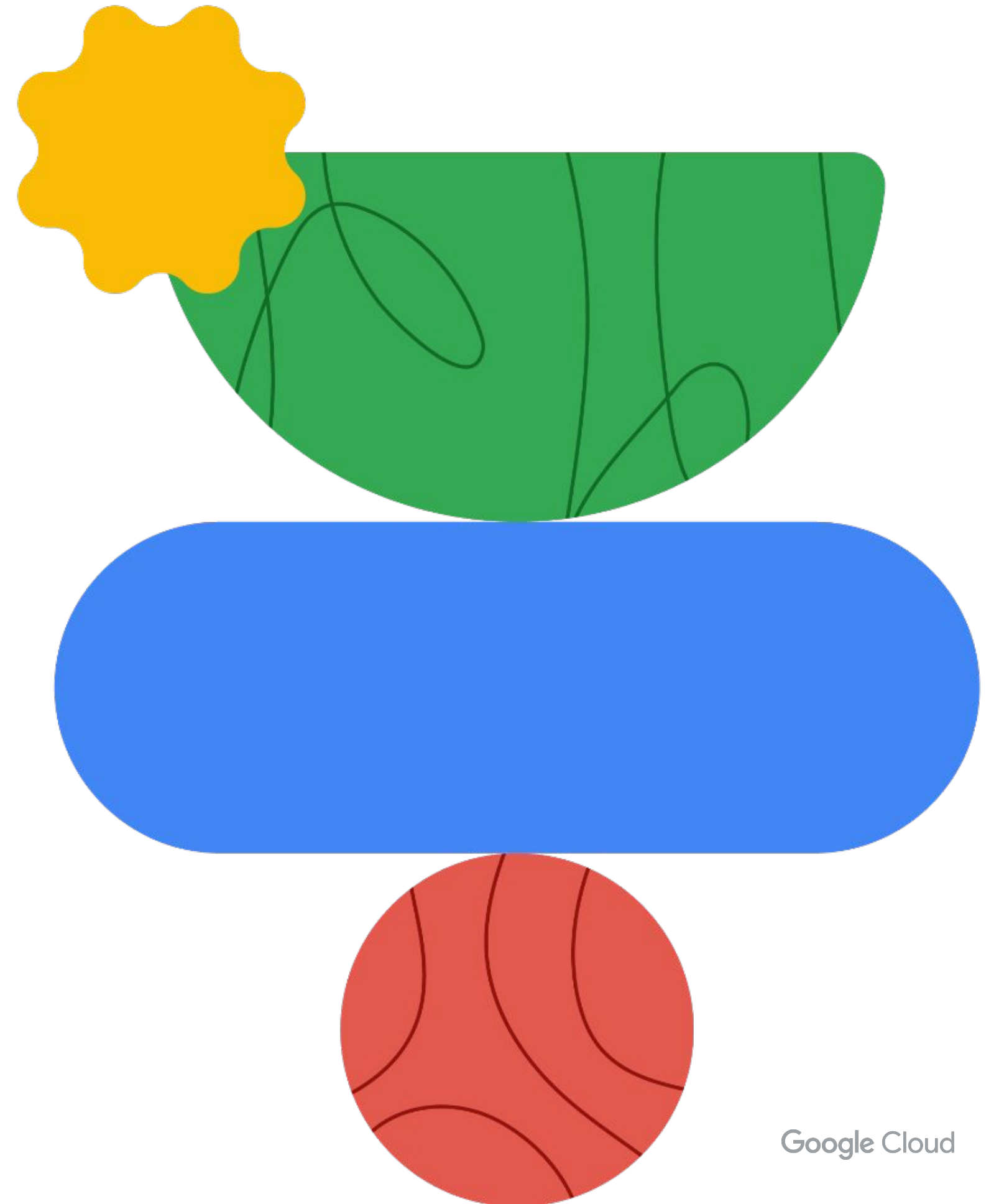
Answer

An organization wants to innovate using the latest technologies, but also has compliance needs that specify data must be stored in specific locations. Which cloud approach would best suit their needs?

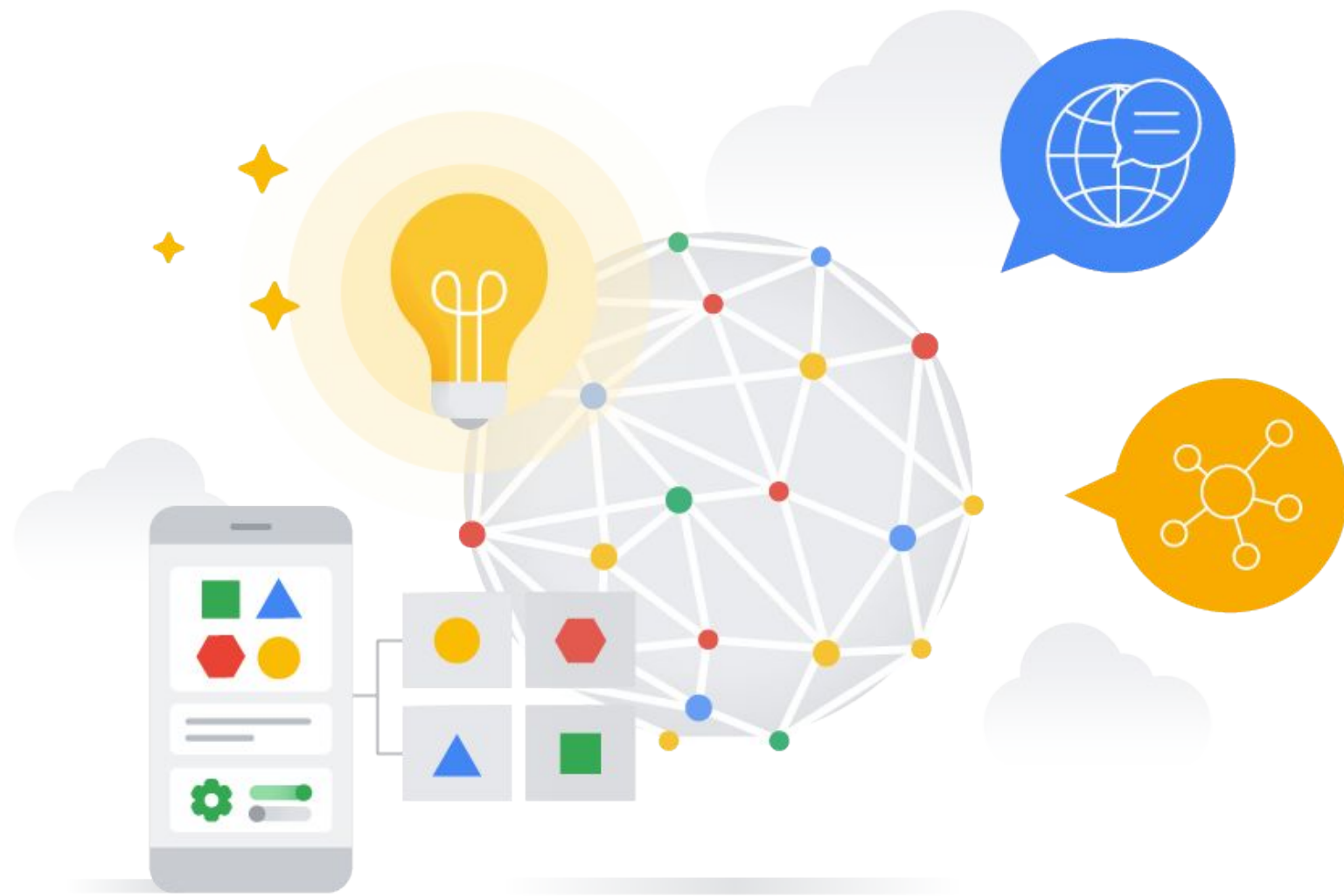
- A. Public Cloud
- B. Multicloud
- C. On-premises infrastructure
- D. Hybrid Cloud



How a network supports digital transformation



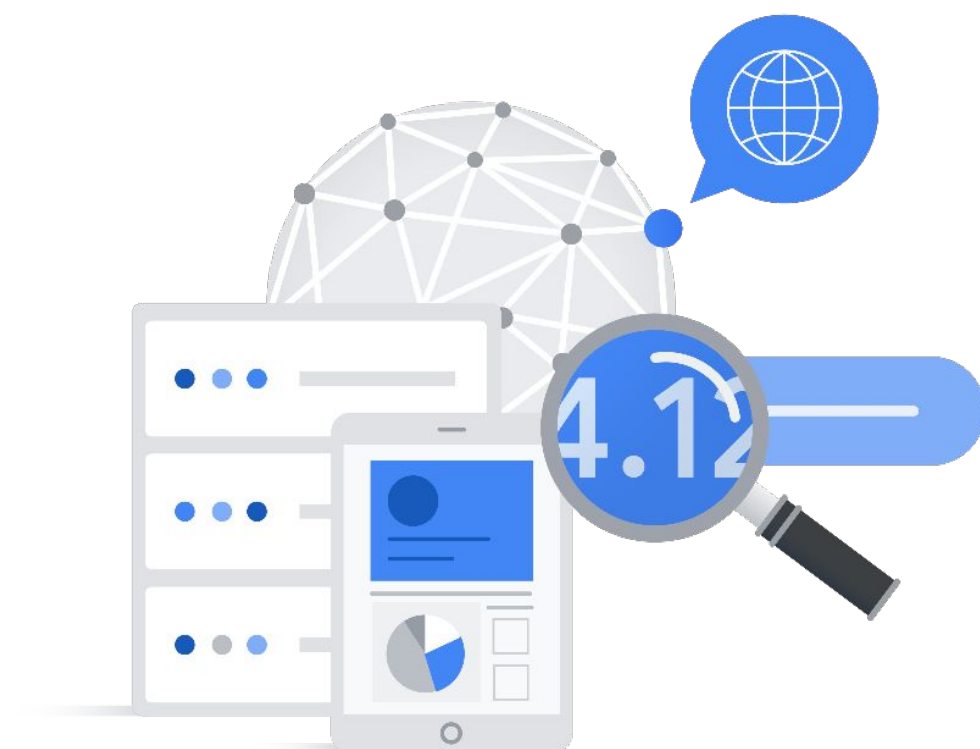
The importance of a reliable network



The ability to connect customers, employees, cloud applications, and devices enables modern organizations to succeed.

With every innovation, the underlying apps and services rely on the network to communicate and connect.

Internet protocols



IP address

Internet Protocol



Domain name



DNS

Domain Name System

IP address



IP stands for Internet Protocol



A series of numbers that can identify a network or the location of a particular device on a network

Domain name



An easy-to-remember name that maps directly to an IP address or set of IP addresses on the internet



The unique name that appears after the @ sign in email addresses and after www. in web addresses



The domain name **example.com** might translate to the IP address **192.168.200.8**

Domain Name System



A DNS server is a database of domain names mapped to IP addresses used by computers to communicate with each other.

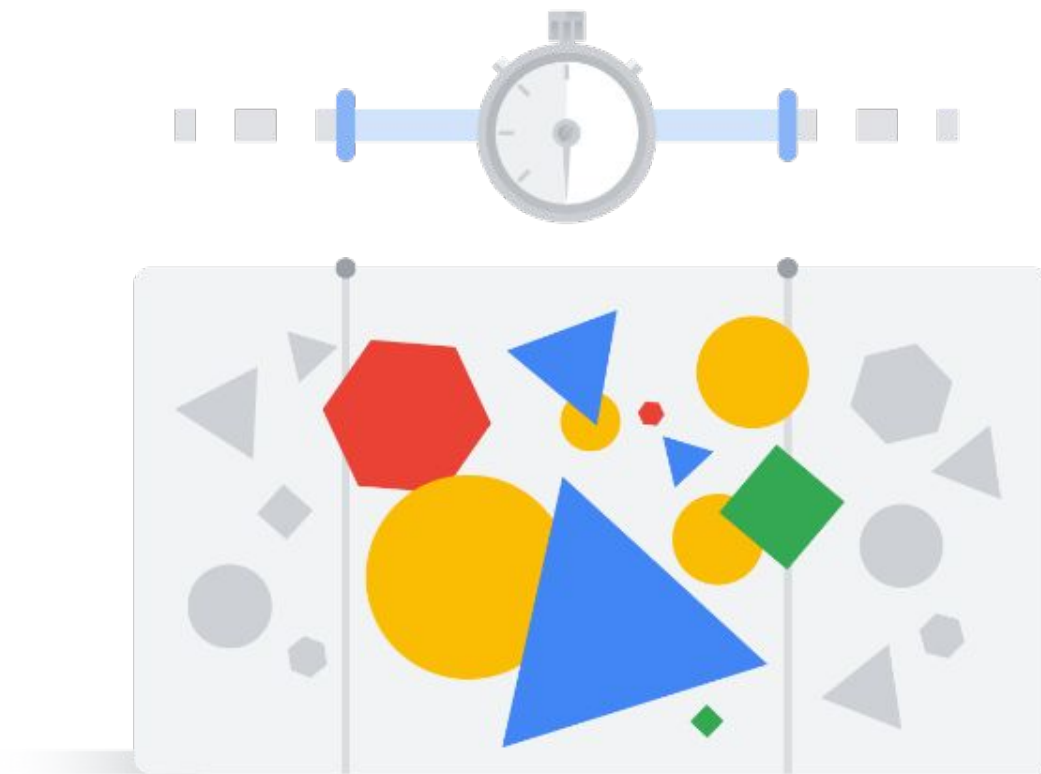


It's like the phone book of the web. Every time you visit a website, your computer performs a DNS lookup.



In the same way a phone book translates a name like "Acme Pizza" into the correct phone number, DNS translates a web address like **www.google.com** into the IP address of the computer hosting that site.

Network performance and measurement



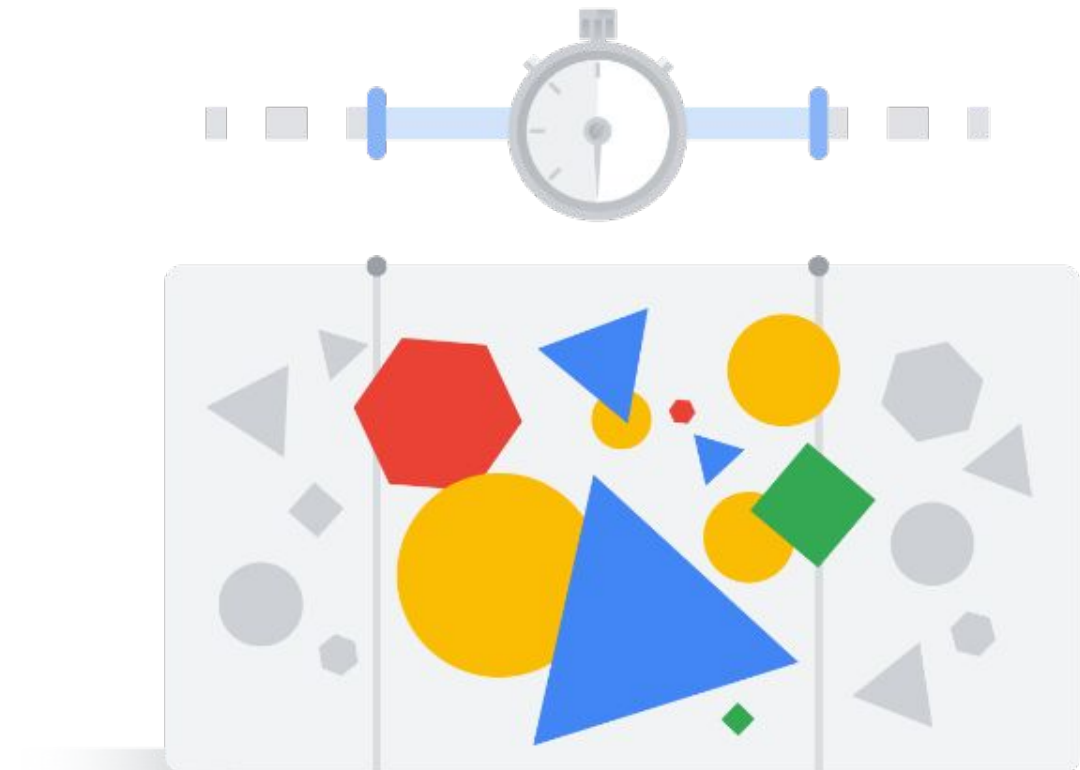
Bandwidth



Latency

Bandwidth

- ✓ A measure of how much data a network can transfer in a given amount of time
- ✓ Measured in megabits per second (Mbps) or gigabits per second (Gbps)
- ✓ Higher bandwidth allows a computer to download information more quickly



Bandwidth

Latency



Latency



The amount of time it takes for data to travel from one point to another



Measured in milliseconds



Describes delays in communication over a network

Quiz

Question

Which network performance metric describes the amount of data a network can transfer in a given amount of time?

- A. Latency
- B. Bandwidth
- C. Domain Name System (DNS)
- D. Fiber optics

Quiz

Answer

Which network performance metric describes the amount of data a network can transfer in a given amount of time?

A. Latency

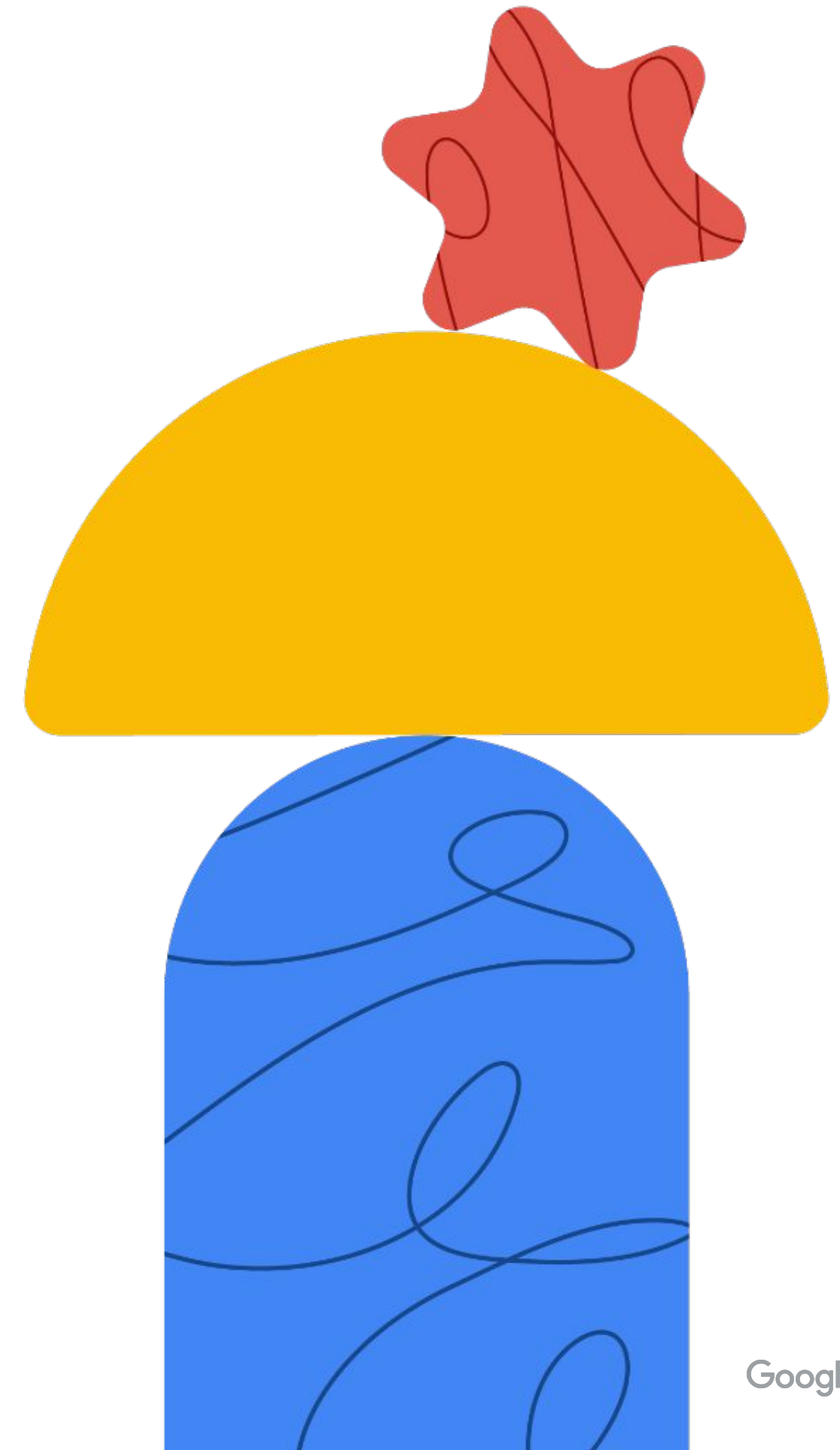
B. Bandwidth

C. Domain Name System (DNS)

D. Fiber optics



Google Cloud regions and zones

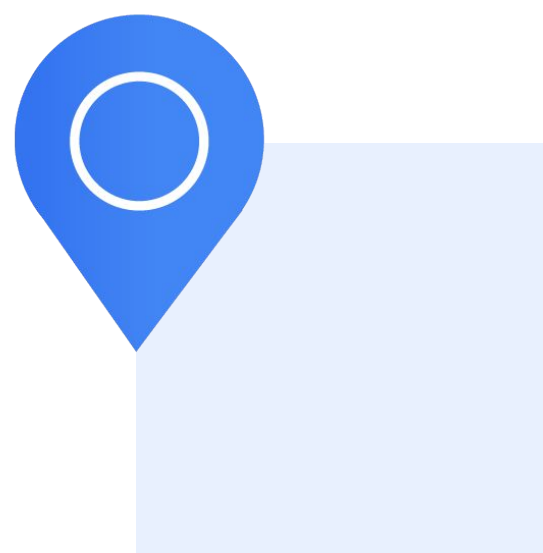


Google's network is one of the largest of its kind

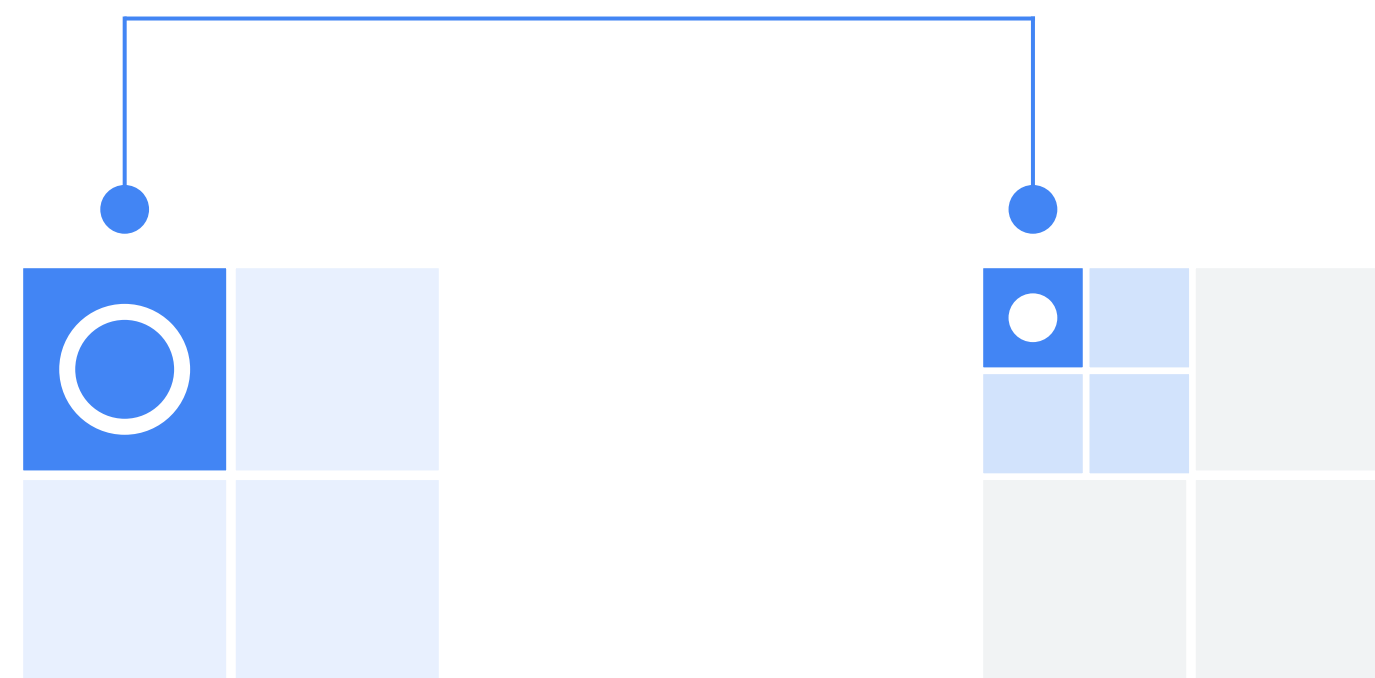


Regions and zones

Regions represent independent geographic areas and are composed of zones.



Location



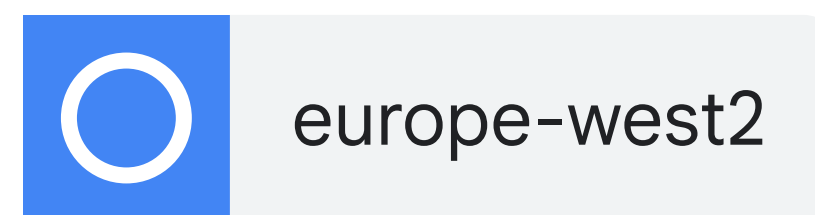
Regions

Zones

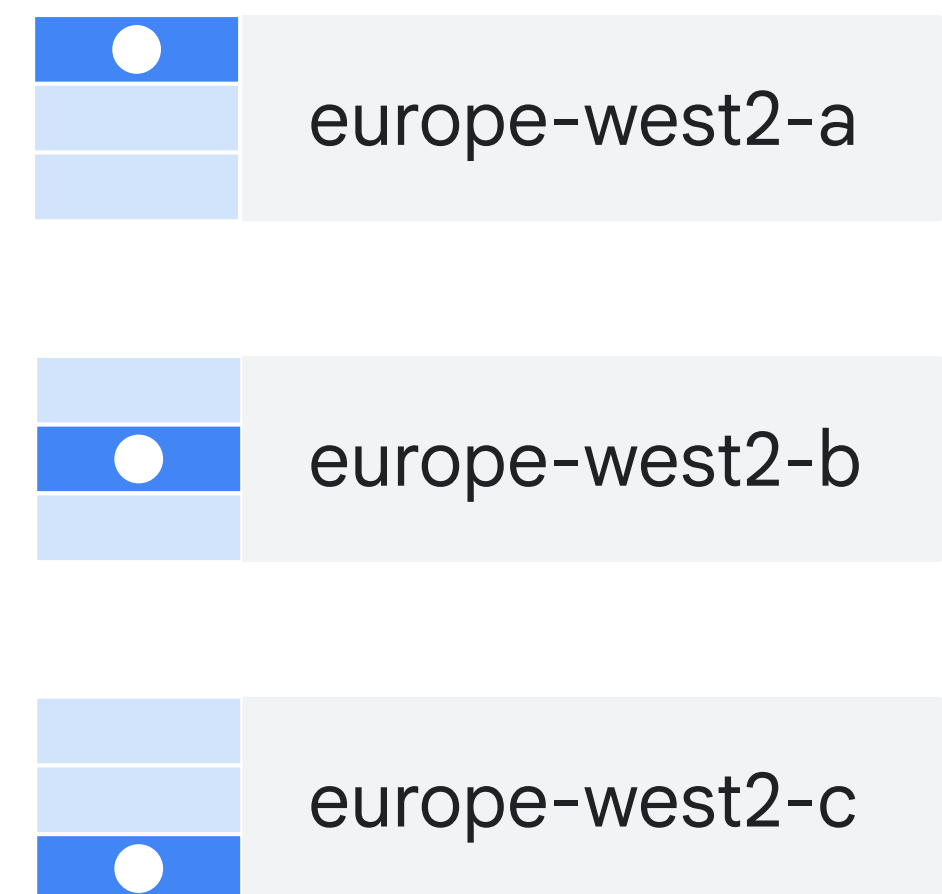
Regions



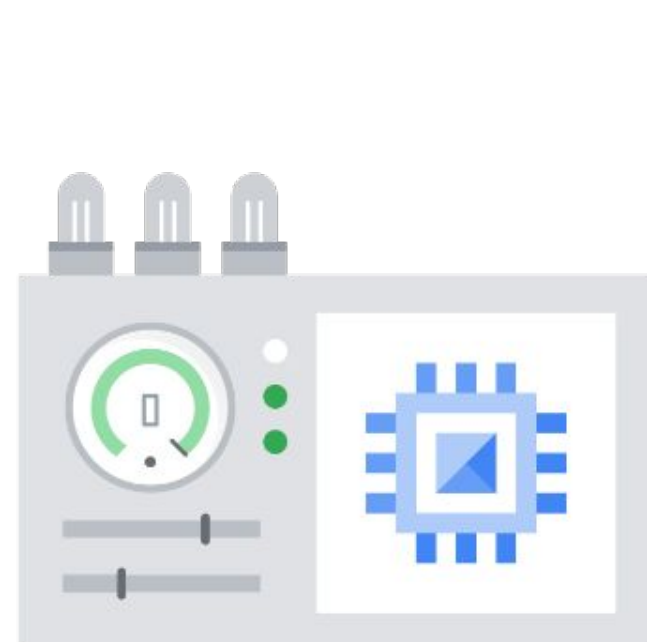
Region



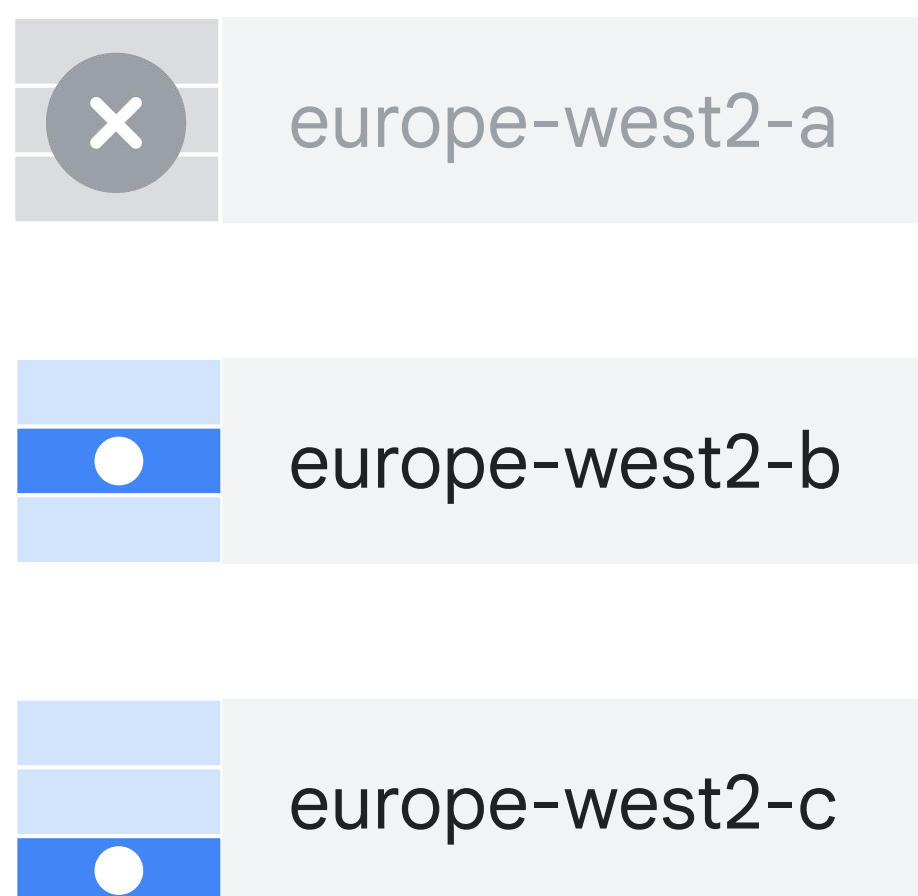
Zones



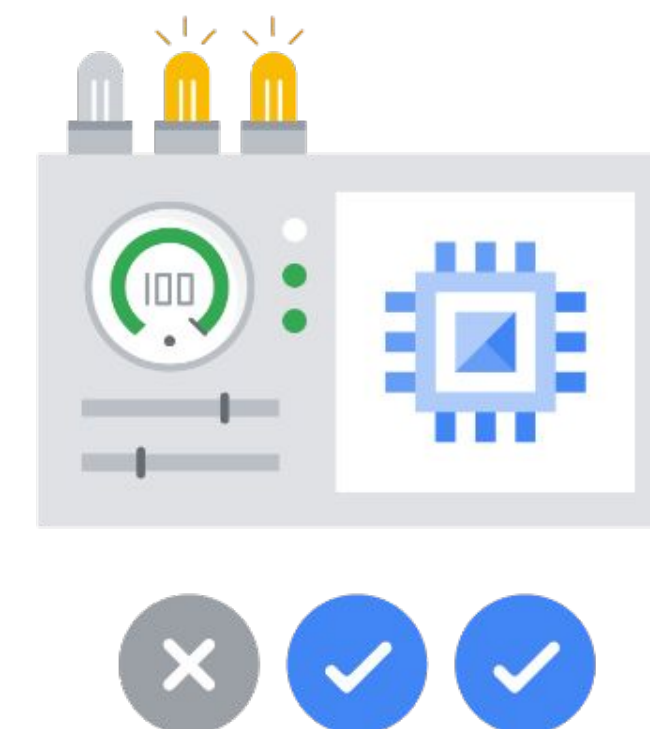
Zones can ensure resource redundancy



Virtual Machine



Zones



Resource redundancy

You can also run resources in different regions

Natural disaster



Region 01

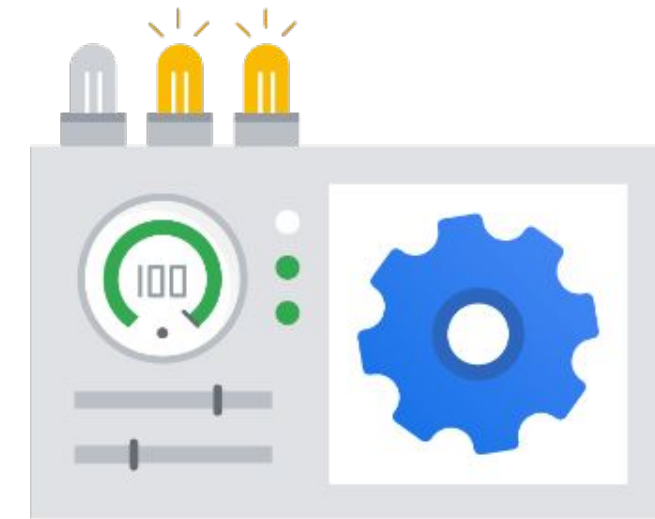


Region 02



Region 03

Regions



Application

Quiz

Question

An organization wants to ensure they have redundancy of their resources so their application remains available in the event of a disaster.

How can they ensure this happens?

- A. By putting resources in the Domain Name System (DNS)
- B. By assigning a different IP address to each resource
- C. Using the edge network to cache the whole application image in a backup
- D. By putting resources in different zones

Quiz

Answer

An organization wants to ensure they have redundancy of their resources so their application remains available in the event of a disaster.

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- C. Using the edge network to cache the whole application image in a backup
- D. By putting resources in different zones



Module 1

Digital Transformation with Google Cloud



Lessons

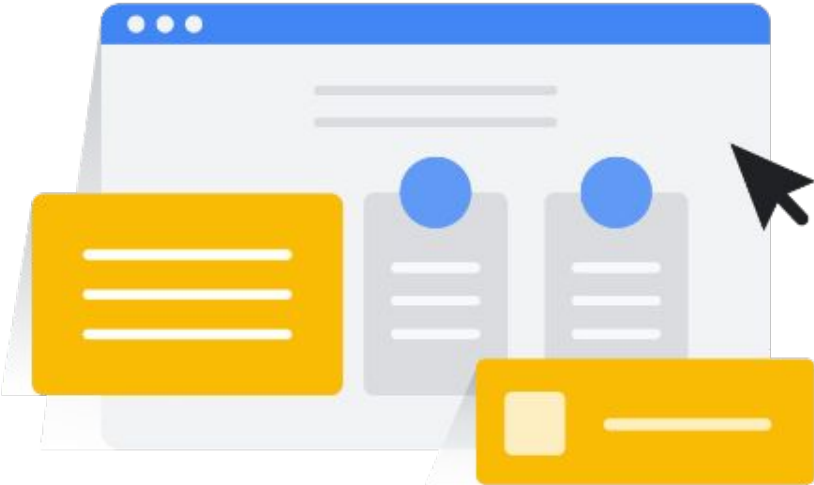
- 01 Why cloud technology is transforming business
- 02 Fundamental cloud concepts
- 03** Cloud computing models and shared responsibility

Cloud computing service models



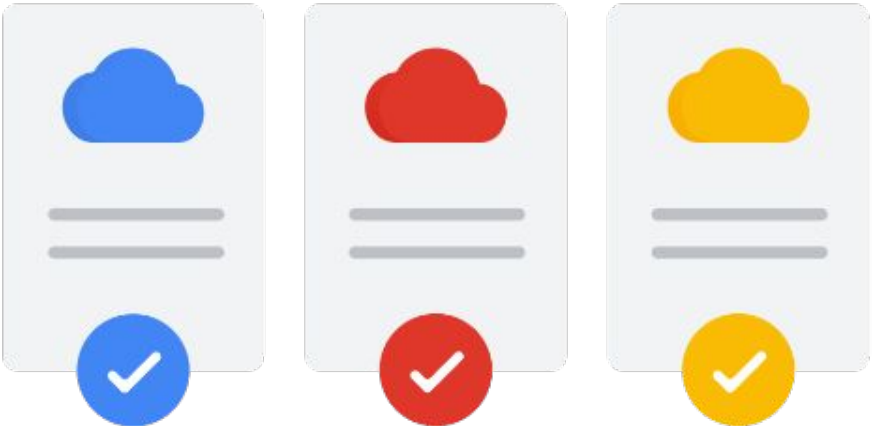
01

Infrastructure as a service (IaaS)



02

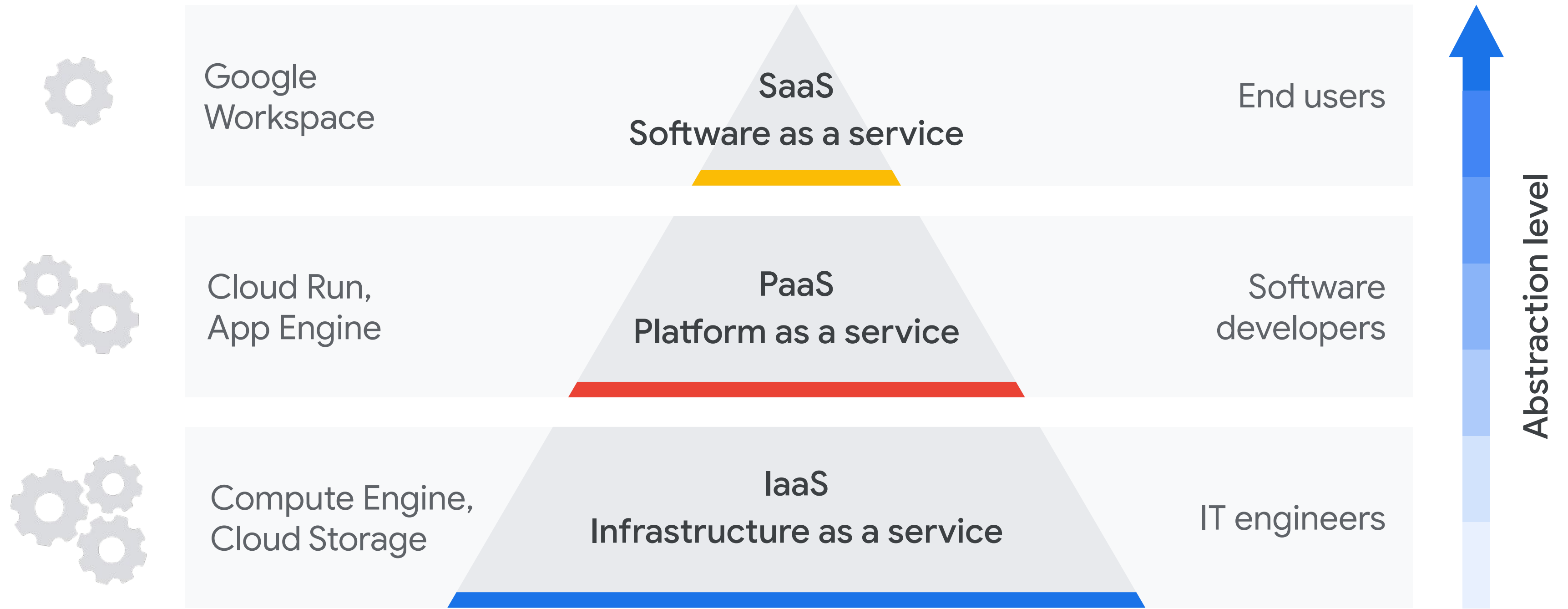
Platform as a service (PaaS)



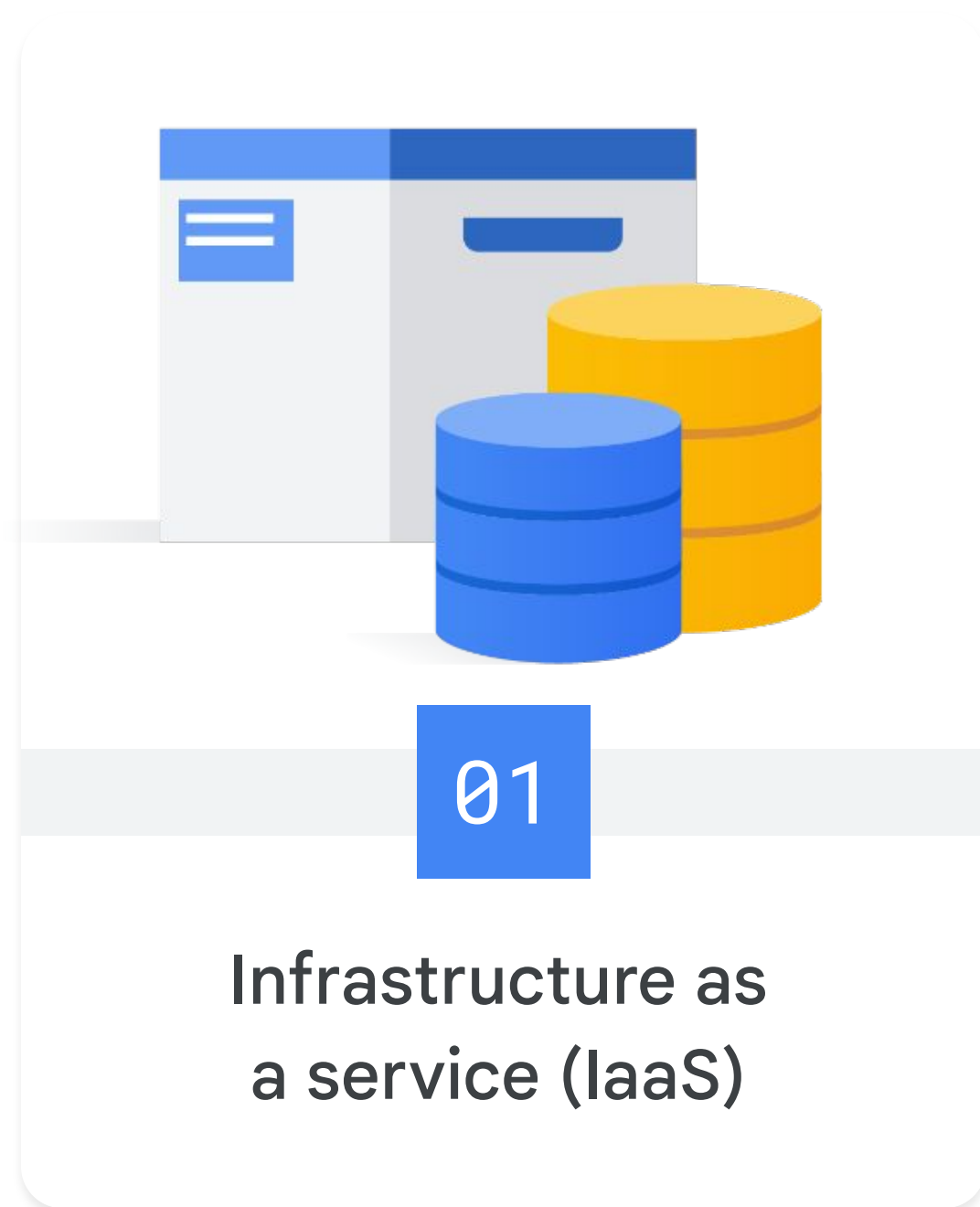
03

Software as a service (SaaS)

Abstraction hides underlying infrastructure



Features of IaaS



- On-demand availability of almost infinitely scalable infrastructure resources over the internet.
- Lease resources instead of having to buy hardware outright, paying only for what's used.
- Same technologies and capabilities as a traditional data center without having to physically maintain or manage it.

The benefits of IaaS



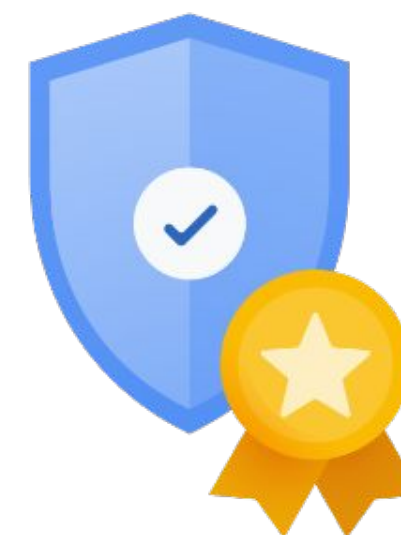
Economical



Efficient



Boosts
productivity

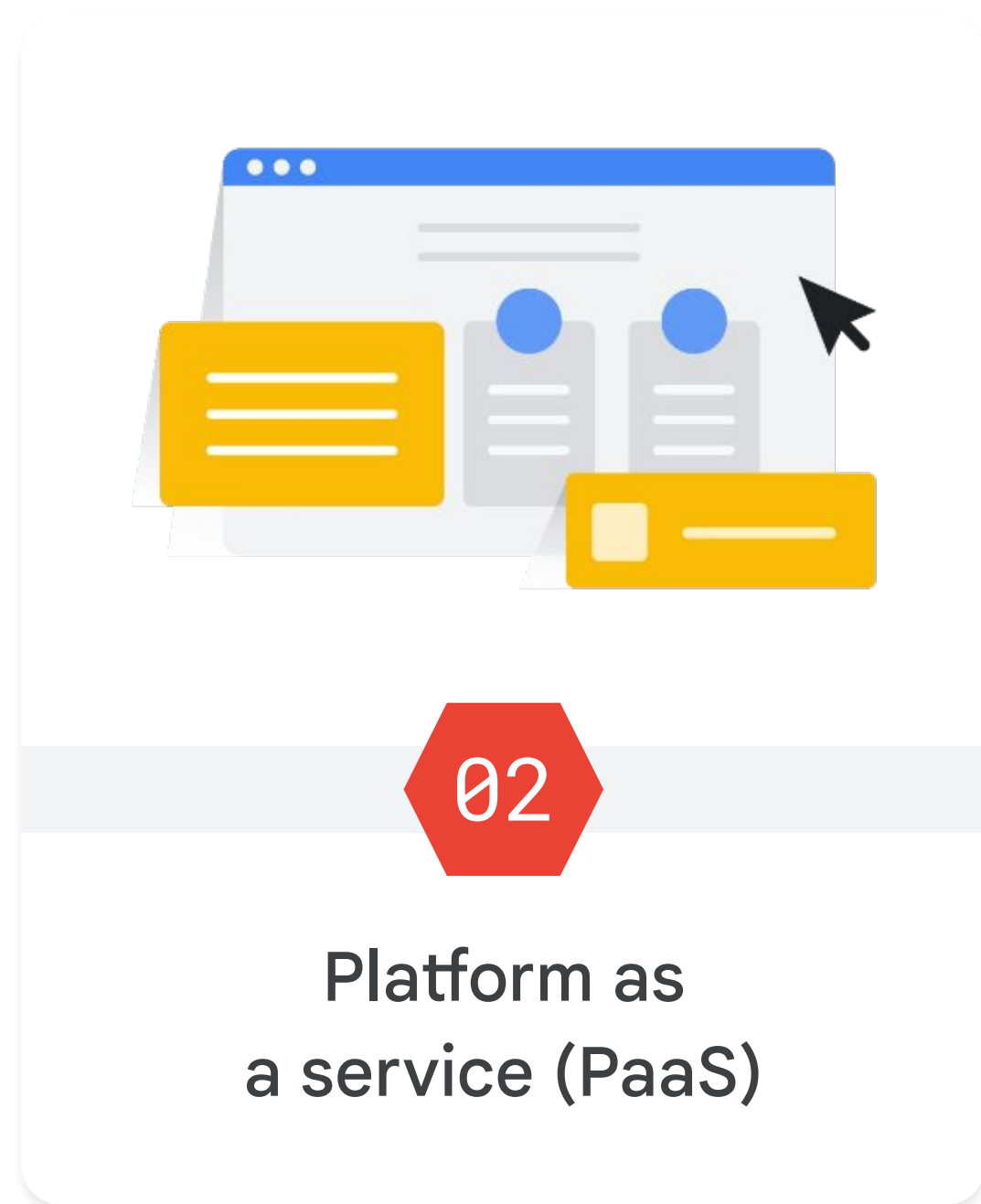


Reliable



Scalable

Features of PaaS



- Provides a platform for developers to develop, run, and manage their own apps
- No need to build and maintain the associated infrastructure
- Can use built-in software components to build applications
- Reduces the amount of code written

The benefits of PaaS



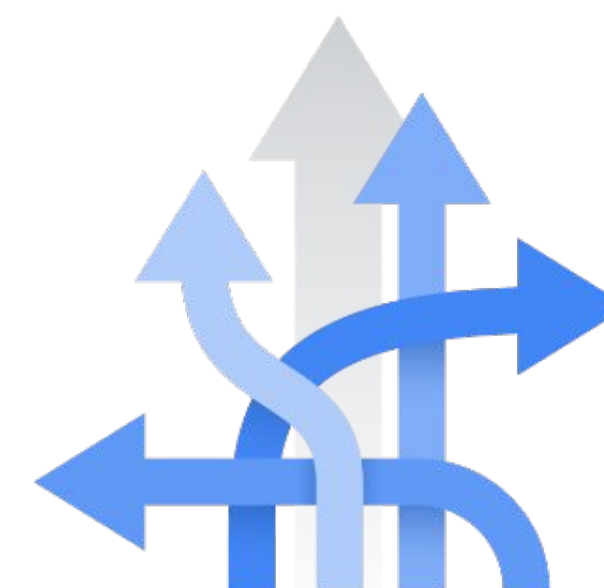
Reduces
development time



Scalable

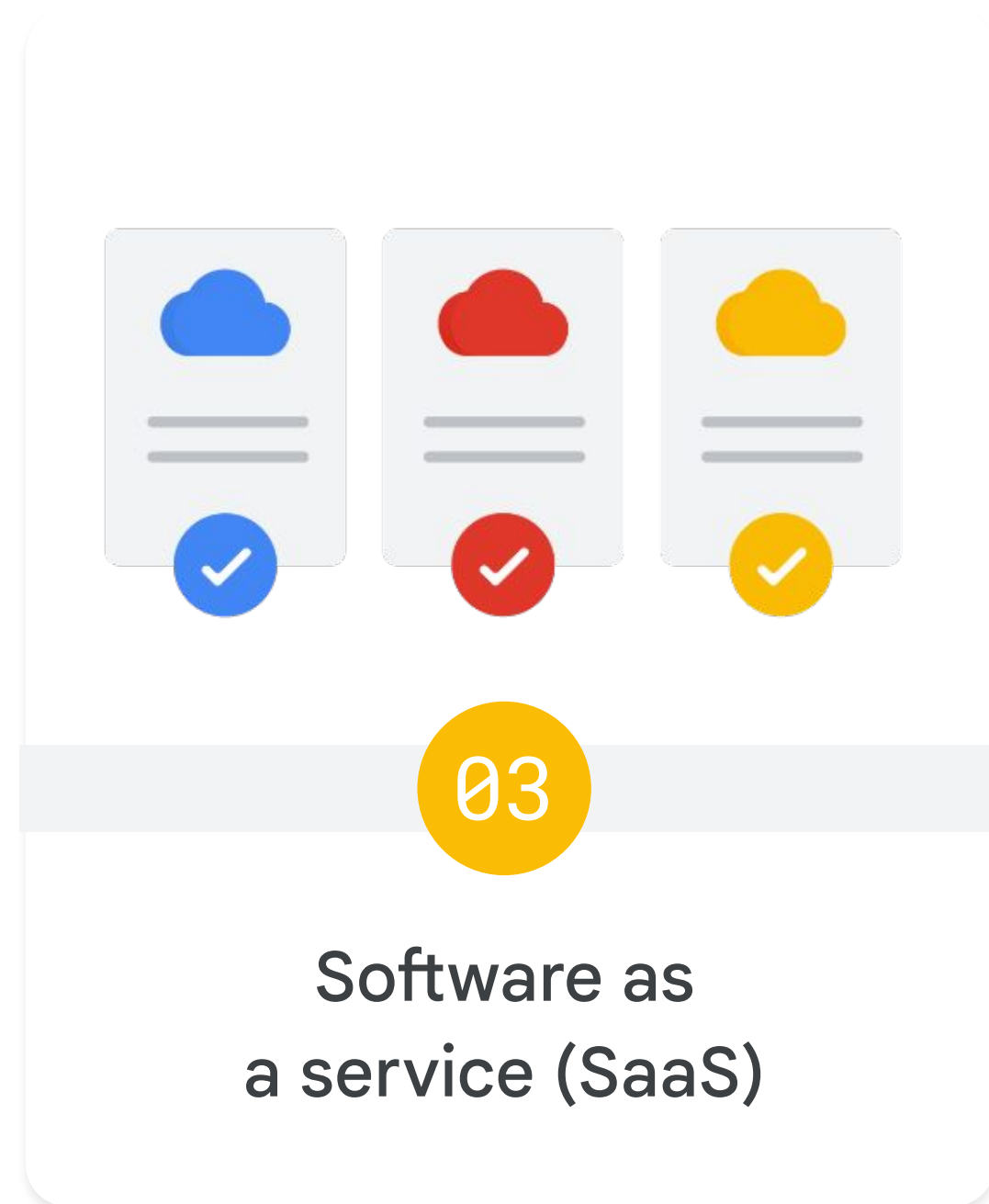


Reduces
management



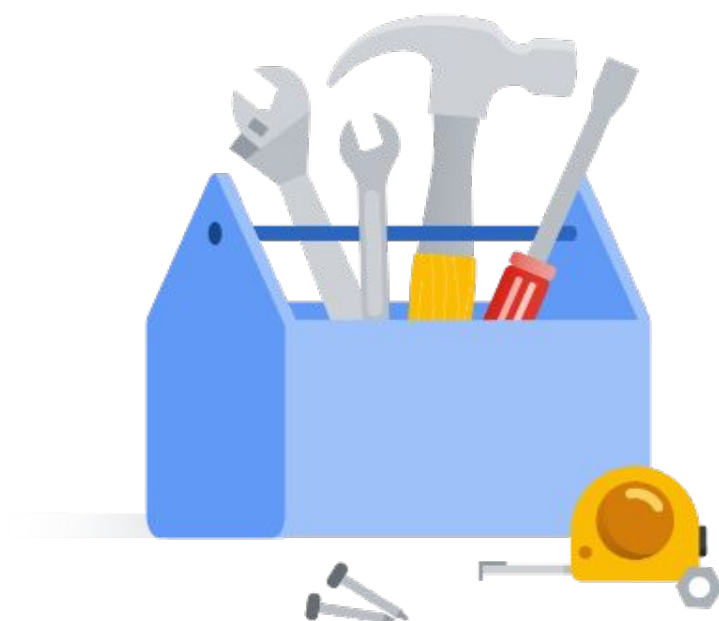
Flexible

Features of SaaS



- Abstracts technology completely from the consumer
- The end user doesn't need to care about the underlying infrastructure
- Organizations pay a subscription fee for access to a ready-to-use software product
- **Google Workspace** is a Google Cloud SaaS product

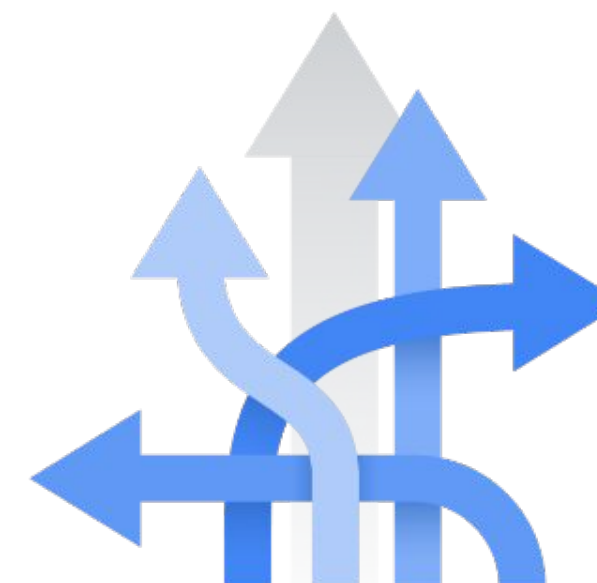
The benefits of SaaS



Low
maintenance



Cost-effective



Flexible

Quiz

Question

Which cloud computing service model offers a develop-and-deploy environment to build cloud applications?

- A. Infrastructure as a Service (IaaS)
- B. Platform as a Service (PaaS)
- C. Software as a Service (SaaS)
- D. Function as a Service (FaaS)

Quiz

Answer

Which cloud computing service model offers a develop-and-deploy environment to build cloud applications?

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- C. Software as a Service (SaaS)
- D. Function as a Service (FaaS)



Quiz

Question

Which option best describes a benefit of Infrastructure as a Service (IaaS)?

- A. It's efficient, as IaaS resources are available when needed and resources aren't wasted by overbuilding capacity.
- B. It reduces development time, as developers can go straight to coding instead of spending time setting up and maintaining a development environment.
- C. It's cost-effective, as all infrastructure costs are handled under a single monthly or annual subscription fee.
- D. It has low management overhead, as all administration and management tasks for data, servers, storage, and updates are handled by the cloud vendor.

Quiz

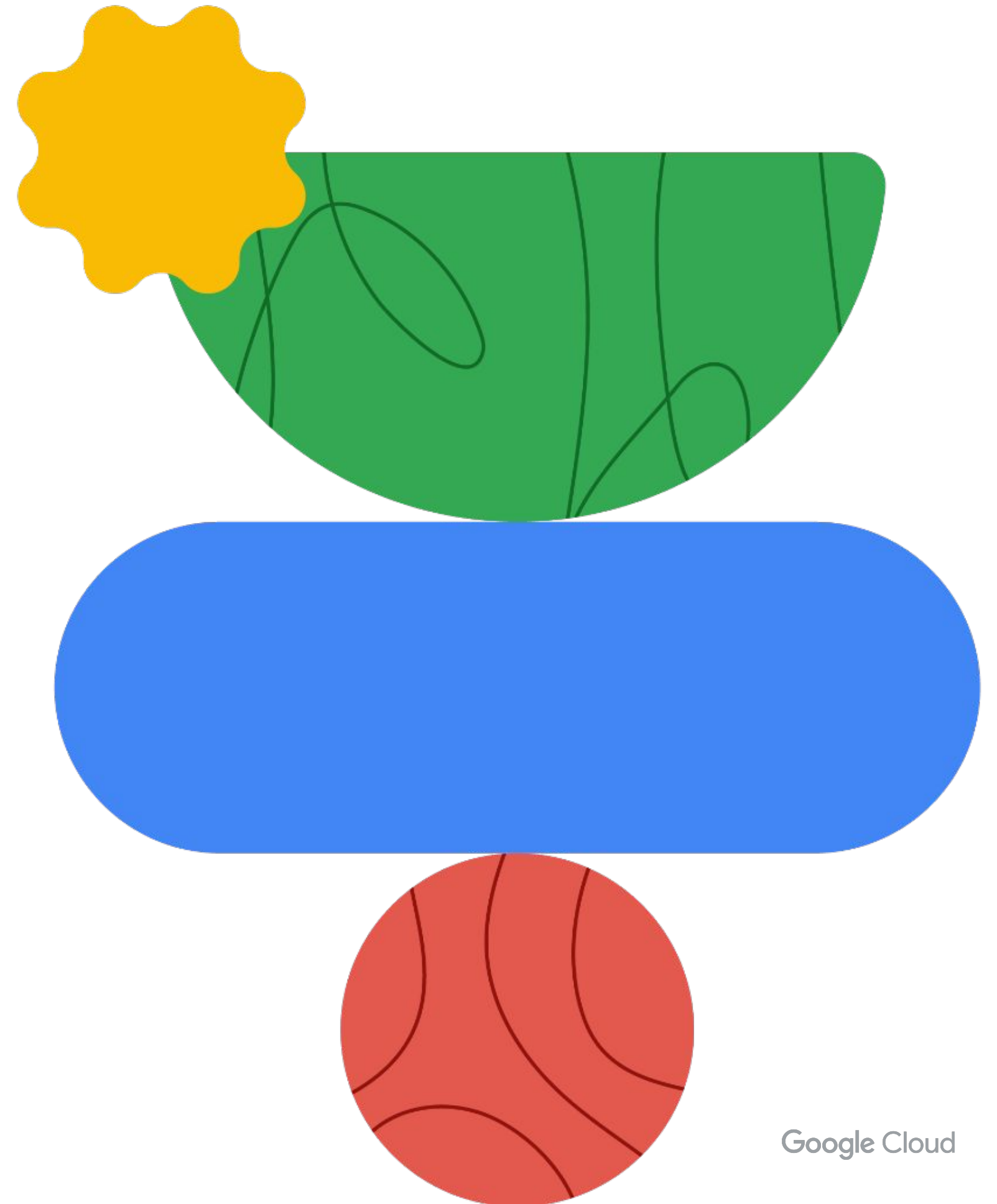
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The shared responsibility model



The shared responsibility model



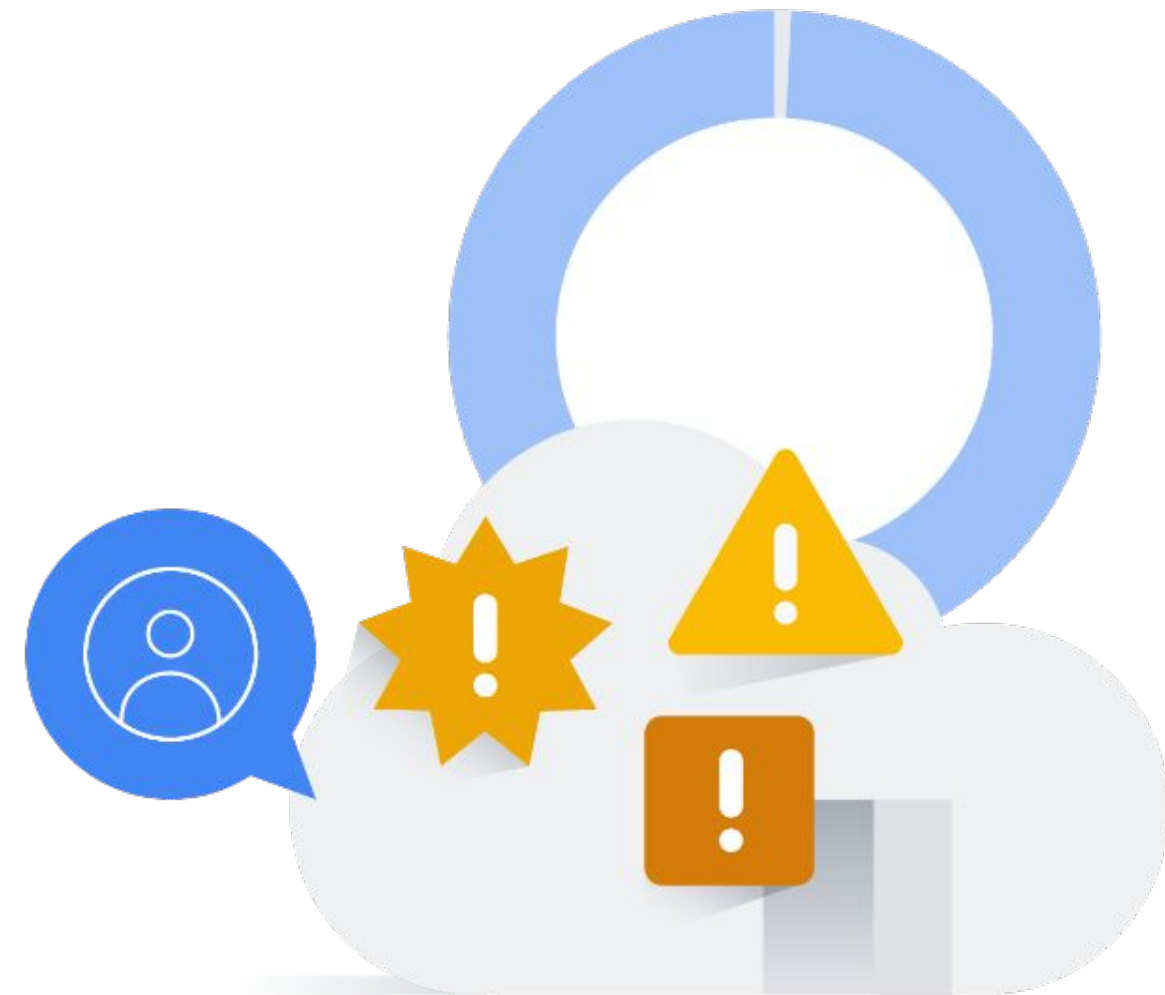
Organization
is always responsible
for securing their data

Cloud provider
is always responsible for
securing the infrastructure

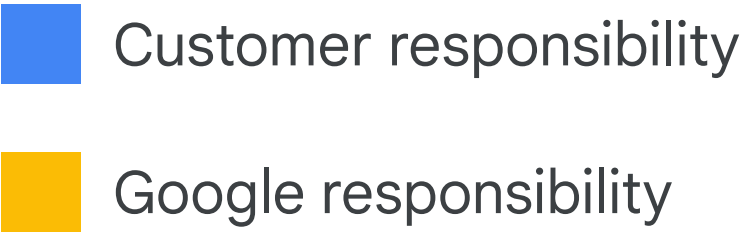
“

99% of all cloud security failures will result from user error

Is the Cloud Secure?
Gartner, 2019



Ratios of responsibility



	On-prem	IaaS	PaaS	SaaS
Content	■	■	■	■
Access Policies	■	■	■	■
Usage	■	■	■	■
Deployment	■	■	■	■
Web App security	■	■	■	■
Identify	■	■	■	■
Operations	■	■	■	■
Access and authentication	■	■	■	■
Network security	■	■	■	■
Guest OS, data & content	■	■	■	■
Audit logging	■	■	■	■
Network	■	■	■	■
Storage + encryption	■	■	■	■
Hardened Kernel + IPC	■	■	■	■
Boot	■	■	■	■
Hardware	■	■	■	■
Physical security	■	■	■	■

Quiz

Question

What is seen as a limitation of on-premises infrastructure, when compared to cloud infrastructure?

- A. The on-premises networking is more complicated.
- B. Scaling processing is too difficult due to power consumption.
- C. Maintenance workers do not have physical access to the servers.
- D. The on-premises hardware procurement process can take a long time.

Quiz

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Quiz

Question

In the cloud computing shared responsibility model, what types of content are customers always responsible for, regardless of the computing model chosen?

- A. The customer is responsible for all infrastructure decisions, server configurations and database monitoring.
- B. The customer is responsible for securing anything that they create within the cloud, such as the configurations, access policies, and user data.
- C. The customer is responsible for security of the operating system, software stack required to run their applications and any hardware, networks, and physical security.
- D. The customer is not responsible for any of the data in the cloud, as data management is the responsibility of the cloud provider who is hosting the data.

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Google Cloud